

DREAMS OF AN ASTRONOMER ❧

By CAMILLE FLAMMARION

Translated from the French by E. E.

FOURNIER D'ALBE ❧ ❧



D. APPLETON AND COMPANY
NEW YORK :: :: MCMXXIII

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I. A VOYAGE IN THE SKY

CHAPTER I

A VOYAGE IN THE SKY

INTRODUCTION

IT WAS at Venice. The lofty windows of the ancient Ducal Palace of the Speranzi opened upon the Grand Canal. The orb of night was mirrored in the waters by a furrow of silver spangles, and the immensity of the sky stretched over the towers and cupolas.

When the musicians borne by the gondolas had turned the corner of the canal to glide towards the Bridge of Sighs, their last choruses vanished in the night, and Venice seemed to go to sleep in that profound silence known to no hive of humanity but the Queen of the Adriatic. This Venetian silence was untroubled save by the cadenced beats of the old clock, and perhaps I should not have appreciated the whole depth of the universal muteness but for the regular oscillation of that apparatus designed for measuring time. The continuous “tick-tock” marked out the silence, and, curiously enough, seemed to intensify it.

Seated in the embrasure of the high window, I contemplated the shining disc of the Moon enthroned in an azure sky filled entirely with its light, and I remembered that this luminary of the night, so tranquil and calm in appearance, moved a thousand yards in space at each beat of the clock. This fact struck me for the first time with a certain force, perhaps on account of the enveloping solitude.

Gazing upon the lunar globe, in which I could distinguish with the naked eye the ancient seas and geographical outlines, I bethought myself that it was still perhaps inhabited by beings organised differently from ourselves who can live in an extremely rarefied atmosphere; but what struck me even more forcibly was its rapid revolution round the Earth, at the rate of 1,000 yards at each beat of the clock, making 38 miles a minute, 2,280 an hour, 53,800 a day, or 1,500,000 miles for each lunar month.

I saw in my mind the Moon revolving round us from west to east in less than a month, and at the same time I felt, so to speak, the daily movement of

the Earth about its axis, also from west to east, which makes the sky appear to move in the opposite direction. While I was still reflecting, indeed, the Moon had actually shifted and descended in the west towards the steeple of the Chiesa. These terrestrial and celestial movements, softer than those of the gondolas gliding on the limpid waters, bear us along through reality as through a dream. They measure the days and the years as we pass, like fleeting shadows, while they endure for ever. The silent Moon, sphinx of the sky, shone already on the waters silvered by her splendour millions of years ago, while terrestrial humanity was still awaiting its slow unfolding in the limbo of future possibilities. Strange animals peopled the forests which covered the continents, fantastic fishes pursued each other in the floods, vampires clove the air, and two-footed crocodiles, which seem to be the ancestors of those of Egyptian mythology, showed themselves in the clearings on the banks of the rivers. Later the same Moon shone on the birth of the flowers, on the nests of the first birds. But how many nights had she not illuminated with pale beams before the first glance from a human eye fell upon her, before the first human thought ascended towards her! To-night she shines upon a populous and active humanity, flourishing cities, marble palaces, built amid the clouds. Just now, at my feet, in a gondola a pair of lovers called upon her to witness their eternal vows, forgetting that her rapid phases are the symbol of our changefulness and our shortness of life. Yes, she has been the confidante of many mysteries, and for a long time yet will radiant youth sing under her sky its eternal song of love. But one day, a poor, enfeebled lamp, she will only shine upon a cemetery of ice; there will be no more clocks for measuring hours nor human beings to count them. Thus I mused, in the bright moonlight which seemed to intensify all the shadows and to deepen all the abysses between the palaces plunging into the black water. This neighbouring world exists at a distance of 240,000 miles from us. Our thoughts fly thither in a flash. With the speed of light, the distance is covered in $1\frac{1}{3}$ second. In imagination I took flight up to the distant luminary. I forgot Venice, the Adriatic, and the Earth, and I felt myself carried beyond the confines of the terrestrial atmosphere.

(1) TWO HUNDRED AND FORTY THOUSAND MILES FROM THE EARTH

I seemed to approach the pale Phœbe and to arrive suddenly above the immense chain of the lunar Apennines, which separate the “Sea of Vapour” from the “Sea of Rain,” not far from the central meridian. I recognised, just as

I had so often observed them in the telescope, the amphitheatres and craters of Archimedes, Autolycus, and Aristillus, and I hovered for some time over the steep cliffs of the “Sea of Serenity.” I saw the traces of old submersions and I distinguished several craters almost obliterated by formidable land-slides. I got accustomed to this view the more rapidly for the fact that astronomical instruments have long familiarised us with this neighbouring world, and that certain details of lunar geography are better known than are many points of terrestrial geography. Those immense amphitheatres, those yawning craters, those steep-walled mountains, those deep valleys, those numerous cracks in the soil—we have studied them all and we know them. We find there the geographical result of considerable volcanic activity, craters 2 miles in depth and 60, 100, or 150 miles wide, mountain peaks 4 or 5 miles high, plains and valleys where the traces of successive selenological epochs are traceable. In the lower depths I observed the effects of a sensible atmosphere, surface changes produced over immense stretches of ground by the action of the Sun’s rays during days fifteen times as long as ours, changes of aspect due to the frost of the long lunar night and the thaw under the midday Sun, long white streaks traversing the circular plains; something like geysers in activity; short-lived plants without any terrestrial analogy—a whole world still alive, apparently in its last death-struggle. My thought and my gaze rested on the pale figure of the Earth’s satellite, and I asked myself whether there was not alive at that moment, in some ancient city at the bottom of a crater or a valley, some thinking being, with its eyes raised to the sky, contemplating the Earth where we are and asking the same question: whether any intelligent beings lived on the surface of that immense globe throning for ever over their heads, and presenting to their minds the same riddle which their abode presents to us.

While I thus reflected about our neighbour in space, the orb of night had sunk in the west, and I saw at some distance from it on the left a star shining with a reddish glow, shedding rays of fire over the heavenly vault. I was not long in recognising in this ardent star our neighbour the planet Mars, and I forgot the moon over this other celestial island, the sister of our own, which has so many analogies with our planet.

Here, said I to myself, is the planet of greatest interest to ourselves, the one we know best. It gravitates round the sun along an orbit traced at a mean distance of 143 million miles from the central luminary. Our Earth passes through its annual revolution at a distance of 92 million miles. There are, therefore, on an average, 51 million miles between the two orbits. On the night of my vigil, Mars happened to be at its minimum distance from the Earth.

Fortunately, as the two orbits are neither circular nor parallel, the real distance is sometimes reduced to 37 million miles. Light, which takes $1\frac{1}{3}$ second to traverse the distance between the Earth and the Moon, takes 200 seconds, or 3 minutes 20 seconds, to cross the celestial abyss which separates Mars from the Earth, It seemed to me that I really spent those 3 minutes in flying the distance, and I entirely forgot the high window of my Venetian palace over the aspect of the new world to which the flight of my thought had brought me.

(2) THIRTY-SEVEN MILLION MILES FROM THE EARTH

It is not very far, astronomically speaking. It is, in fact, quite near, a few paces away. The world of Mars is the first station of the solar system, the first planet we meet on leaving the Earth to visit the remote regions of the heavens. The farther we move away from the Earth, the smaller grows the apparent size of our own world. Seen from the Moon, our planet hangs in the sky like an enormous moon, four times the size of our own satellite, and sixteen times as luminous, for it is isolated in space and reflects the light received from the Sun, as is done by the Moon and the various planets of the solar system. From about 250,000 miles, therefore, the Earth still appears of a considerable size, being about four times the size of the full Moon. At $2\frac{1}{2}$ million miles it appears ten times smaller in diameter, but still shows a perceptible disc. At the distance of the orbit of Mars, at the time when the planets are in greatest proximity (37 million miles), the Earth no longer shows a sensible disc, but is still the biggest and brightest star in the entire heavens. The inhabitants of Mars, therefore, admire us as a brilliant star in the sky, showing aspects similar to those which Venus shows to us. We are their morning and evening star, and no doubt their mythology has erected altars to us.

When I arrived on that planet, it was about midday on its central meridian. I noticed two small moons revolving rapidly in their sky, and I alighted on the slope of a mountain overlooking a distant sea. The sea was shallow and full of water-plants. The panorama reminded me of that which one sees from the terrace of the Nice Observatory, and I seemed to see a Mediterranean of calm water, of a rather dark bluish-green colour. But it was a different element, and I saw that the plants were of a species unknown on Earth. Airy navies consisting of a sort of bird-fishes glided through the atmosphere, and I soon found that the inhabitants of this celestial territory have received by natural evolution the enviable privilege of flying through the air, and that their method of locomotion is particularly aviation. Gravity is feeble on the surface of the

planet, and hence the density of beings and objects on that planet is much less than it is with us. Engineering science has for many centuries reached a high degree of perfection. They have carried out immense works, incomparably superior to those achieved on our planet during the last century, and they have transformed their globe by gigantic operations which earthly astronomers are just beginning to appreciate by means of the telescope. One may easily understand, indeed, that that world should be more advanced than ours, because it is more ancient chronologically, and because, being smaller than our globe, it has cooled down more rapidly and has run through the phases of organic evolution at a greater rate. Its years are nearly twice as long as ours, in the proportion of 365 days to 687. While we count 37 years on Earth, the Martian only counts 20, and a man of 79 years on Earth is only 40 Martian years old. This is an advantage of 88 per cent. Its condition of habitability, its climate and meteorology, its days and its nights, are analogous to ours. Even from where we are we can observe its continents, its polar snows which melt in the spring, its canals which also change with the seasons, its humid plains periodically varied by vegetation, its clouds, generally very light, but dense enough towards the polar regions, its mists in the mornings and especially in the evening, above all, the perpetual changes, incomparably more intense than those of the Earth's surface—in a word, all those manifestations of an activity greater than that of our own home of the present day.

I only delayed on Mars for the time necessary to form a general idea of the life which animates our neighbouring globe and to make sure that it is more active than that of terrestrial humanity, and I found myself, some moments later, transported to the annular world of Saturn.

(3) AT SEVEN HUNDRED AND FIFTY MILLION MILES

The conception of time, the appreciation of duration, are essentially relative to the state of our mind. If we sleep profoundly for seven or eight hours, that time will have made a gap in our life of no greater length than that produced by ten minutes of sleep. The miners who by the collapse of a shaft are entombed for five or six days before being rescued, always believe that they have not been cut off for more than twenty hours. Buried on a Tuesday, for instance, they will not believe that they have had to wait till Sunday. On the other hand, one may seem to pass several hours, very slowly, in a dream of a few seconds. A friend of mine told me that one day, as he was riding through a wood, his horse bolted and threw him into a ravine. He said his fall had

certainly not taken more than three seconds, but that during those three seconds he had passed in review at least ten years of his life in all their successive details and without any apparent hurrying of events. Then, again, who has not observed how long the minutes may seem during some hours of waiting?

The orbit of the Earth round the Sun being 92 million miles, and that of Saturn 888 million, there are 796 million miles between the two orbits. Light traverses this gap in 70 minutes. My fancy flew this distance with the speed of light, and I was aware of these 2,240 seconds required to cover the distance at the rate of 186,000 miles a second. Yet I am sure that I did not spend all that time in traversing the distance to Saturn, nor even the lesser time corresponding to the distance between Mars and the ringed planet, for the first stroke of ten had sounded on the old clock when I forgot Mars and fixed my attention on Saturn, and I arrived at my destination before the hour had finished striking.

I alighted on the tenth satellite, whence one can easily appreciate the grandeur of the Saturnian system. The enormous planet of, a diameter more than $9\frac{1}{2}$ times that of our globe, with a surface 90 times that of the Earth, and a bulk 745 times that of our floating home, is surrounded by gigantic rings measuring 178,000 miles across. Girt by this multiple ring, the planet presides over a retinue of ten satellites revolving round it in a system having a radius of 8 million miles, a system which in itself constitutes a universe larger than that known to the ancients. Until the age of truth inaugurated by the conquests of modern astronomy, nobody on our planet, no poet, no philosopher, no thinker, had guessed the real grandeur of the proportions on which the universe is constructed. How small our Earth appears seen from the Saturnian system! It is barely seen, once in six months, as a small luminous dot near the sun, shining for a few minutes in the evening after sunset, or a few minutes in the morning before sunrise.

(4) TWO THOUSAND FIVE HUNDRED MILLION MILES FROM THE SUN

In the depths of space, at a distance from the sun more than 30 times our own, under a glow of light and heat 900 times feebler than that which we enjoy, there roams the world of Neptune, among conditions of life quite different from those which obtain on Earth. Those short-sighted naturalists

who affirmed even quite recently, with professorial emphasis, that the abysses of the ocean are condemned to an eternal sterility, because the conditions of light and pressure are absolutely different from the conditions near the surface, have received from Nature herself the rudest contradiction which can be inflicted upon the pedantic science of pretenders to infallibility. This contradiction, however direct and absolute, has not discouraged them all, for there are still some who declare that life can only exist on worlds having conditions identical with ours. Always the reasonings of the fish who affirms—quite sincerely—that it is impossible to live outside water! Let us leave these teachers to their illusions and continue our ascent. Astronomy must be the great teacher of philosophy.

The distant world of Neptune, on which every year equals nearly 165 of our years, and where ten years represent the whole historical interval which separates us from the Romans (we must remember that 1,650 years ago the Romans reigned at Paris and in Gaul, and neither France nor any of the present-day nations were thought of), this neighbouring world, I say, is well fitted to teach us to enlarge our narrow and personal conceptions, especially as regards the measurement of time. The calendar of that planet is just as exact, just as precise, as ours, and a Neptunian year is not longer to those slow and reflective beings who inhabit the place than is a terrestrial year to those hurrying and agitated persons who swarm in our turbulent cities. Yet a Neptunian adolescent of 20 has really lived nearly 3,300 terrestrial years, without knowing that such a time is called “very long” by the inhabitants of our planet, whom such a life would carry back to the epoch of Homer and ancient Greece.

It would be impossible even with the most careful examination to discover any point of comparison between the beings which live on the Neptunian world and those which we know on Earth. None of our classifications, whether of the animal kingdom, vast and diversified though it be, or of the vegetable kingdom, highly complex in itself, could be applied to them. It is another world, absolutely different from this one. Spectrum analysis indeed establishes the fact that its chemical composition is quite other than that of our terrestrial home. The organisms which live on the surface of the different planets are the resultant of the forces acting upon them. The origin of the human form lies in the ancestral forms of the long animal series whence it has gradually emerged, and of which it is the highest perfection, and these primitive animal forms go back in an unbroken chain to the rudimentary organisms unprovided with the senses which are the glory of man, organisms which inaugurated the

manifestations of life, but which can hardly be described as living. They are neither animals nor plants. They appear to be organised substances, already distinct from the inorganic kingdom, but as yet only simple chemical combinations endowed with a sort of diffused vitality, an elementary protoplasm, the germ of all developments of terrestrial life, both animal and vegetable. The first organised beings were formed in the bosom of the warm waters of the oceans which covered the entire surface of the earth at the time when the geological periods began. Their intrinsic nature, their properties, their faculties, were already the resultant of the chemical composition of those waters, of the density and temperature of the surrounding medium; the variation of this medium and of the condition of existence have brought about corresponding changes in the development of this genealogical tree, and, according to the habitat of the organisms, whether in the deep, middle, or upper regions of the waters, on the sea-shore, in the low-lying plains, on sunny slopes or mountain-tops, the genealogical tree gave rise to more and more diversified organisms. Present-day terrestrial humanity is the last flower, the last fruit of this tree. But all this life is terrestrial from root to summit, and on every planet the tree is different. Life is Neptunian on Neptune, Uranian on Uranus, Saturnian on Saturn, Sirian on the system of Sirius, Arcturian on that of Arcturus, appropriate to every medium, or rather, more strictly speaking, produced and developed by each world according to its physical state and in harmony with that primeval law which all nature obeys: the law of progress.

This immense symphony of life, adapted to every world according to conditions of space and time, develops like a universal choir, the parts of which are separated from each other by deserts of space and by eternities of time. It appears to us discontinuous because we can only hear one note at a time. But in reality there is no absolute separation either in space or time. Jupiter will not be inhabited by thinking beings for millions of years to come; from the point of view of the Absolute, the interval is not greater than that which separates yesterday from to-day.

All this happens and accomplishes itself naturally, and as if God did not exist. And indeed the being whom the inhabitants of the Earth have hitherto defined as God does not exist. The Buddha of the Chinese, the Osiris of the Egyptians, the Jehovah of the Hebrews, the Ormuzd and Ahriman of the Persians, the Teutates of the Gauls, the Jupiter of the Greeks, "God the Father" or "God the Son" of the Christians, or the great Allah of the Mussulman, are human conceptions, personifications invented by man in which he has

embodied not only his highest aspirations and his sublimest virtues, but also his grossest prevarications and ugliest vices.

Man has conceived as God in his own likeness. It is in the name of this pretended God that monarchs and pontiffs have in all the ages and under cover of all religions bound humanity in a slavery from which it has not yet freed itself. It is in the name of this God who “protects Germany,” “protects England,” “protects France,” “protects Italy,” “protects Russia,” “protects Turkey,” protects all the divisions and all the barbarities, that even in our own day the so-called civilised people of our planet have been armed in war against each other and, like mad dogs, have hurled themselves upon one another in a conflict over which falsehood and hypocrisy, seated on the steps of the thrones, figure a “God of Armies” as presiding, a God who blesses the daggers and plunges his hands in the smoking blood of victims to mark the foreheads of kings. It is to this God that altars are raised and Te Deums are chanted. It is in the names of the Gods of Olympus that the Greeks condemned Socrates to drink the hemlock; it is in the name of Jehovah that the high-priests and Pharisees crucified Jesus. It is in the name of Jesus, himself become God, that fanaticism ignominiously condemned to the stake men like Giordano Bruno, Vanini, Étienne Dolet, John Huss, Savonarola, and so many other heroic victims; that the Inquisition ordered Galileo to belie his conscience; that thousands and thousands of unfortunates accused of witchcraft were burnt alive in popular ceremonies; that Ravaillac stabbed Henry the Fourth. It was with the express benediction of Pope Gregory the Thirteenth that the butchery of St. Bartholomew drenched Paris in blood and the free thinkers of the reformation were chased out of France; it is for the extermination of supposed heresies that many thousands of brave people have been burnt alive; it was with cross in hand that the peaceful natives of America were savagely massacred by the Spaniards; it was in the names of the gods worshipped in Rome that the Christian martyrs suffered the most awful tortures; it was in the name of the Christian God that the fanatics, led by Bishop St. Cyril, stoned the beautiful and learned Hypatia, and that in later times the Bishop of Beauvais led the virgin of Domremy to the stake; it was in the name of the Bible that the kings of God’s “chosen people” savagely exterminated their neighbours; it was in the name of Allah that the standards of Mahomet covered Europe with armies of assassins and that even now millions of fanatics are ready to rise against the Europeans on the cry of a “Holy War,” that Mahomet the Second painted the walls of St. Sophia with the blood of his steed, that Genghis Khan and Tamerlane marked their paths of conquest with pyramids of severed heads;

it is to the glory of these imaginary deities that even now so many and useless souls condemn themselves to strange penances in the convents, that the Russian stropzi mutilate themselves, that the howling and dancing Dervishes writhe in mad contortions, that certain sects kill their babies and drink their blood. Religious wars have been the most horrible and odious of all, and the most insensate. People have killed each other for the sake of a word or its interpretation, for the sake of an adjective, for the “consubstantiality” of the Father and the Son of the Trinity, for “homoousios” against “homoiousios,” for a thousand other crotchets placed above the most elementary reason and proclaimed articles of faith in the name of a God! This symbol of the oppressions of peoples, of murder and robbery, this infamous being, does not exist, and has never existed.

In making a god in their likeness, as miserable as themselves, men resemble monkeys, who, raising themselves to the idea of God, would figure him as a Grand Ape, dogs who would make him a Grand Dog, fleas who would represent him in the shape of a Grand Flea. But it does not follow that, because this inferior god does not exist, therefore the universe goes on without thought, without a destiny, and without laws. If the believers of all religions are in error, those who deny the existence of any intellectual principle in the world are equally mistaken. Yes, the evolution of nature takes place without the acts of an anthropomorphic deity, without malice and without miracles. The only name which would fit God would be the Unknowable. God remains hidden by the very perfection of nature’s mechanism. In a healthy body we do not feel the passage of the blood through the heart, nor the circulation of blood in the brain, nor that of the air in the lungs, nor the liver, nor the kidneys, nor the stomach, nor the bowels. The attention is not directed to those organs unless they work badly. The world is so arranged that they appear to work by themselves, and that is its divine quality. Everything functions regularly by means of a perfect construction, the gearing of which is invisible and silent, but which we can only judge by the infinitesimal fraction of which we form a part, and the author of which is a transcendental thought, impenetrable to mankind. Force governs matter; *mens agitat molem*; but Thought guides nature.

Yes, the supreme being is unknowable. The human mind cannot comprehend the infinite, eternal, immutable spirit, the organising power of that All of which the Earth and Man are but particles as imperfect as they are mediocre. This Infinite is to the infantile deity imagined by man as the midday sun is to the muddy obscurity of a mole-burrow under the roots of meadow

grass. His existence is proved by the universal organisation. Everything is organised, from the humblest leaf to the world system. An invisible, immaterial element of a spiritual nature, as yet imperfectly revealed by our means of investigation, manifests itself within us and around us. This spiritual principle should be revered as enveloping the world and enfolding us. But the clergy of all religions, of all times and of all countries, have always monopolised the idea of God and appropriated it to their exclusive and intolerant purpose of domination. When they speak of God, they mean their own God. The free spirits who do not acknowledge their figure-head are treated as atheists and hated and persecuted as such. They will not admit that one may be a Deist and yet anticlerical. But they are not too unintelligent to know that it is an injustice, a stupidity, and a lie to treat as atheists those thinkers who deny the divinity of Jesus. The people who dare to reduce God to their size and even to put Him in their pocket are the greatest blasphemers.

It is strange that Man, still in a coarse, savage, and barbarous state, hardly emerged from the primitive shell of ignorance, incapable of knowing even his own body, hardly able to spell the great book of the universe, should have considered himself capable of describing God. He does not know his own little ant-heap and he pretends to discover the unknowable. At a time when nothing was known, when astronomy, physics, chemistry, natural history, and anthropology were as yet unborn, when the feeble and meandering human mind was still surrounded by illusions and errors, human audacity conceived the so-called religions and the gods placed at their heads.

That Confucius, Buddha, Moses, Socrates, Jesus, or Mahomet should have striven to give to mankind a code of morals destined to deliver them from barbarism and to teach them the idea of good, such efforts and achievements cannot but receive the homage and admiration of all who value the intellectual and moral progress of humanity. That the founders and organisers of religious rites should place at the head of every cult an ideal, an inviolable being in whose name they pretend to govern, even that can be recognised as a work of social utility, of a value not rising above the worldly standard, and having no object beyond the general good of man and societies. But that those gods invented by man should be considered as really existing, and in an absolutely imaginary heaven which perished in the first conquests of astronomy, that they should have been, and are still, adored by a certain portion of the human race, and that even in our time legislators of all nations dare to base their politics on divine right, to show the “finger of God” in the most monstrous plagues of the social body, and decorate their battle-flags with a local providence, as in the

time of Joan of Arc, of Constantine, or of David—that is the shocking anachronism, a mixture of imposture and credulity, of hypocrisy and stupidity unworthy of the era of sincere and positive research in which we live, and which should load the functionaries who live at the expense of such a system with the contempt of every independent man.

Definitions are misleading. Were pagans like Socrates, Plato, Aristotle, Marcus Aurelius, Plotinus, not more spiritual than Pope Alexander the Sixth or Cardinal Dubois, who were true atheists?

The search for the nature of the First Cause—I do not say the “knowledge of God,” which would be an expression worthy of a “theologian” and absurd in itself—but simply the search for the Absolute Being, for the origin of the energy which sustains, animates, and governs the universe, for the intelligent force which acts everywhere and perpetually through infinity and eternity and gives rise to the appearances which strike our eyes and are studied by our science—this *search*, I say, could not be undertaken nor even properly conceived before the first discoveries of astronomy and modern physics, that is to say, before the investigations of Galileo, Kepler, and Newton. It is only two centuries ago that the purely religious idea, free from idolatries, from all sorts of mythologies, of errors and superstitions produced by primitive ignorance—it is only during the last two centuries that it was possible for this idea to arise out of modern scientific evolution. All the religions existing at the present day have been founded during ages of ignorance, when we knew nothing about the earth or the heavens. True religion, i.e. the union of free spirits in the search for truth, can only be the work of an epoch like ours, in which some courageous and disinterested spirits free from the hypocrisy of false doctrines, yet without falling into the puerile atheism of superficial minds which only see the outer shell, will sincerely and freely apply all branches of science to the search for the intimate constitution of the universe and of the human being. The future will teach us. To-day we know but little; we are only beginning to learn. The unknown God conceived by the thinkers, by Socrates, by Plato, by Marcus Aurelius, by Voltaire (as ardent a Deist as he was a violent anticlerical), by Newton, by Descartes, by Linné, by Euler, by Spinoza, by Kant, by all pure Deists, surpasses in his grand immensity all the poor inventions of the clergy of all denominations. One cannot see the creator of the hundred million suns of the Milky Way looking down upon a small village in Judea and inspiring Judith to seduce Holophernes with the object of cutting off his head after betraying him with her caresses; or conferring on Joshua the power of arresting the movement of the solar system to give him time to

exterminate the besiegers of Gibeon! What sort of opinion had such writers of the Supreme Being? And what opinion of him is still held by those preachers who continue to teach this “Holy Scripture”?

The Infinite cannot be comprehended by the Finite.

He who has made the tour of the world, who has visited Europe, Asia, Africa, and the Americas, reasons in a manner wider beyond comparison with the state of humanity than he who has never left his country. Between the narrow, incomplete, and false ideas of the latter and the judicious, exact, and just appreciations of the former there is the difference of night and day.

Unfortunately we preach to the deaf. For one man who reasons there are a hundred who do not. The struggle against the domination of the spiritual directors is very platonic and the clergy despise it. The Church has organised marriage, birth, and death into ceremonies which seduce the imagination and please the women. Compare the civil marriage and the religious marriage; the former cold, dull, insipid; the latter impressive and attractive with the altar garlanded with flowers, and the enchantment of the music which makes the creator descend into the bosom of the spouse, “*Veni, Creator Spiritus.*” It will be centuries before the religious form of marriage is entirely replaced by the civil. A certain free-thinking father refrained from having his four boys baptized so as to leave them entire liberty of conscience. All four got baptized on the eve of their marriage as their brides wished to be married in church. Faith or convention, that is how the world goes—and the priests smile at the simplicity of the layman.

What is Sunday for most Christians but a day for fine clothes?

Tradition has created a distinguished society often permeated with hypocrisy, but to belong to which is “good form.” Ancient errors are preserved without being credited. Convention governs the “well-disposed” people.

Independence of spirit is the rarest of phenomena. All religions are sacred and respectable if they raise our thoughts to a higher ideal, when they console the afflicted and relieve misery. But let them not be exploited, and let there be no killing in their name! Ideal and sentiment are part of the domain of thought, with as much right as Reason. It is a mistaken policy to suppress them, and it plays into the hands of reactionaries, who profit by the errors. To claim that science demonstrates the non-existence of God and of the soul is an unscientific argument. An education without ideals or responsibility, which neglects conscience and proclaims rights without duties, is as false as that of

the Catechism which teaches the creation of Adam and Eve, the temptation of the serpent, the universal deluge, the incarnation of God, the Virgin Birth, the resurrection and ascension of Jesus to the right hand of God the Father, or the resurrection of our bodies (articles of faith which cannot, without heresy, be interpreted as symbols, but must be taken literally), and it is socially more dangerous, as we see by its fruits for the last thirty years, the gradual increase of crime and the rule of the Apaches and anarchists of all sorts. Why not follow the way of enlightened wisdom? If the socialists were permeated with spiritual truth, they would hold the world; they would continue the work of their predecessor, Jesus Christ, in the light and with the positive methods of modern science. The false gods invented by man, the legends, the superstitions, the errors, lies, and hypocrisies, are not necessary to secure a place in the educational system for the sense of honour, of duty, of justice, and of personal conscience; and this is often forgotten by modern educators who have suppressed everything without putting anything in its place.

Let us have no sectaries of any kind!

Humanity grows up. We are no longer children.

At the distance from the Earth which suggested such reflections, the distance of the planet Neptune, the farthest limit of the solar system known at present to astronomy, our judgment on the works of man is quite different from that which satisfied us before we left our country. We contemplate the solar system in all its grandeur, we recognise the smallness of our own little planet compared with the vast space in which it moves, and the short time of its revolution round the Sun, and we feel that our ordinary terrestrial estimates have hitherto been based upon those narrow and limited sentiments circumscribed by the horizon of the church tower. We free ourselves from them and find ourselves in a position to judge the immensity of creation with greater liberty, independence, and integrity. But far as Neptune is from our terrestrial home, it still forms, like ourselves, part of the solar family. Other planets still unknown to terrestrial astronomy gravitate beyond Neptune, the first of them probably at a distance 48 times as great as the distance between the Earth and the Sun, that is to say, at 7,500 million miles, in an immense orbit which it takes at least 330 years to accomplish. The celestial voyage which I have begun takes me beyond the outermost regions of the solar system. Flinging myself into the infinite heavens, I arrived at another system by penetrating into the cosmic domain of a star.

(5) AT TWENTY-FIVE BILLION MILES

Every star is a sun shining by its own light. The Sun which illuminates us has 1,300,000 times the volume of the Earth and weighs 333,000 times as much. The dimensions and the masses of the stars are of the same order. A large number of them are much more voluminous and their masses are still more considerable.

Whatever star we approach, we find in it a sun like a blinding furnace. These innumerable centres of light, heat, electricity, and gravitational attraction only appear to us as small luminous points on account of the immense abysses which separate us from them. The nearest sun, our nearest star in space, burns at 276,000 times the distance which separates us from the Sun, i.e. 25 billion miles from here.

Travelling with the speed of an express train flung into space at 40 miles an hour towards the nearest star without any stoppage or any slowing down, we should not arrive at our destination until after an uninterrupted flight of 75 million years.

Travelling with the speed of the swiftest projectile which the most ingenious man-killers have yet constructed, a speed which we can reckon as double that of sound, or 2,200 feet per second, we should yet require a million and a half years to cover that distance.

If that star were to burst with a terrific explosion, and if the noise of the catastrophe could be transmitted to us at the ordinary speed of sound in air, we should not hear that explosion until three million years had elapsed after its occurrence.

We should see the star shining steadily in the sky for four years after the catastrophe which had destroyed it, because light travels through space with a speed of 186,000 miles a second, and it would have to travel with this constant velocity more than four years before reaching us.

Seen from that distance, our brilliant Sun is reduced to the rank of a simple star. The planets which gravitate around it, the Earth, Venus, Mars, Jupiter, Saturn, and their brothers of the solar family, are crowded up against it by the perspective of the distance and are invisibly lost in its rays.

Considered at that distance in the sidereal universe, these provinces of the solar empire are recognised as insignificant even by the most optimistic spirit. Even if they did not exist at all, the suns of infinite space would none the less

shed their rays of life and light all around. Our planet, which to us seems so important, becomes a microscopic point impossible to discover by means of senses such as ours, and its history told at that distance becomes like the flight of a dragon-fly or even less, since we should never suspect its existence if we did not know it. It is at such moments especially that the pretensions of Pontiffs and the dogmatic assurance of their adepts show forth in all their absurdity.

I felt transported into the system of that star, the nearest of all those whose distances have been measured, a star belonging to the constellation of the Centaur; it is the Alpha of that constellation. This system is curious and more interesting than ours. Instead of a single sun corresponding to that which shines upon us, two twin suns gravitate one round the other in a time equalling 81 of our years, and separated from each other by a distance of 2,000 million miles. These twin suns are both of considerable brightness (first and second magnitude, seen from here), and greatly superior to the central hearth of our own system. Planets circulate around each of these luminaries under their protecting wings, and receive from their radiation the sources of their fertility and their life. They are illuminated by two different suns, sometimes united in the same sky, sometimes separated and alternating, differing in magnitude and brightness according to the variation of the distances in consequence of the revolutions of these worlds round their respective centres.

These are very different conditions of existence from those which govern the destinies of the Earth and of the planets of our group. Two suns! What curious alternations of seasons! What variations in the climates! What transformations in the doubtlessly very rapid changes of their vitality! What complications of their calendars, in the succession of their years, their summers and winters, their days and nights! The sole fact of the existence of such a system, relatively near to us and already well-known to terrestrial astronomers, testifies to the infinite variety disseminated in the starry depths of the cosmos.

What multiplicity of manifestations of the diverse forces of nature must have been produced in this wealth of solar development—manifestations strange to the phenomena studied on our planet, and which are doubtlessly felt and appreciated by means of senses differing absolutely from those existing in terrestrial organisms, senses awakened, determined, and developed in those distant worlds by their own natural forces.

On worlds illuminated, heated, and regulated by two suns life can only have appeared and organised itself in forms very different from those on Earth, having no doubt an alternating double life, served by other modes of perception, other organs, and other senses. The thinker, the astronomer, the physiologist, can no longer regard terrestrial life as the type of all life. All we could learn, study, or know on Earth will never be more than an infinitesimal and absolutely insufficient part of the immense reality embodied in the innumerable creations of the Infinite.

Yet it is a point which must be insisted upon before pursuing our terrestrial investigations further, that whatever may be the variety of stellar systems, the differences of volume, temperature, density, illumination, electrification, movement, chemical constitution, etc., of the various globes which people the immensity of the universe, all these worlds are linked amongst themselves by the same invisible and imponderable Power which combines them all in a network of extreme sensitiveness. The prodigious extent of the distances which separate these systems one from the other does not prevent their being connected together by some sort of maternal link. The distance from the Earth to the Moon is 240,000 miles. The Moon acts constantly upon all the molecules of our globe and upon the entire Earth, and every one of us weighs a little less when that body shines over our heads than when it is on the horizon. The distance from the Sun to the Earth is 92 million miles; the Sun makes our planet move with a speed corresponding to that distance, and the Earth in its turn displaces the Sun in the heavens. The distance from the Sun to Neptune is over 2,500 million miles. The central globe acts upon that distant world and makes it revolve round it, and on the other hand Neptune makes the Sun revolve round their common centre of gravity, which is at a distance of 144,000 miles from the centre of the Sun. Jupiter displaces the Sun by 460,000 miles, and Saturn by 25,000 miles. The Moon disturbs the Earth to the extent of 2,900 miles. At the same time Jupiter acts upon the Earth, the Earth upon Venus, and so on. On account of this reciprocal influence of all the heavenly bodies upon each other, not a single point can remain in repose for an instant, and no heavenly body can ever come back to the place it previously occupied. All that we call matter is in perpetual motion under the irresistible power of invisible, intangible, and imponderable force.

We have here a fact of capital importance, the consideration of which must always be associated with the conception we can form of the real nature of the universe. We have seen just now that the distance which separates our Sun

from the star Alpha Centauri is 25 billion miles. But this distance is traversed by gravitational attraction. In reality the two suns are not absolutely separated.

They know each other, they feel each other's attraction, and they feel the attraction of all the other suns of infinite space. They both roam about, our own Sun with a speed calculated at about 200 million miles per annum and Alpha Centauri with a speed of approximately 400 million. The other suns of which we know the distance and the movement rush on with similar speeds. Some of them fly with incomparably greater velocities, which attain 200 miles a second, 11,000 miles per minute, 600,000 miles an hour, 15 million per day, and 5 or 6 thousand million miles per annum, veritable starry projectiles of the heavenly fields.

But our whole sidereal universe itself is moving with its hundred million stars through the immensity of infinite space. The movements which we measure are relative and not absolute.

Our sun and its companions are driven through space by some initial force and by the combined attraction of the innumerable stars of our visible universe. Whether this force of attraction is a property inherent in every atom of matter, whether these theoretical atoms by which we explain the appearance of matter in order to account for observed phenomena are centres of force, mathematical points of concentration, or nodes and crossings of ethereal vibrations and undulations, the fact which dominates our analytical contemplation of the universe is that the innumerable worlds which people space are not isolated from each other, but are united by a perpetual and indestructible link.

Here we have a new and important conception of the unity of nature. And what is equally worthy of attention is that this sort of communication between the worlds cannot be defined better than by the word "attraction."

Attraction is therefore the supreme law among the worlds, among atoms, and among beings. The stars which gravitate in the depth of space, the Earth which revolves in the solar rays, the Moon which raises the tides on the surface of the ocean, the molecules of stone or iron which cling together by molecular attraction, the plant which pushes its roots into the nourishing soil or raises its stem in response to light, the flower which turns towards the Sun, the bird which flies from branch to branch seeking a place for its nest, the nightingale which with incomparable song charms the sweet mistress of the night, the man whose heart is troubled at the appearance of a beloved being,

the sound of a beloved voice, or a fond memory—all these beings, all these things obey the same law, that of universal attraction which in diverse forms governs all nature and guides it—whither? Towards yet another attraction, to the attraction of the unknown!

Amid the ignorance of the Absolute which surrounds us in spite of the manifold, courageous, and persevering efforts of science, the fact of the existence of such a force uniting all worlds together must be appreciated at its proper value. It would be impossible to exaggerate its importance. Let us then not forget it: the worlds are in mutual communication by means of attraction.

(6) AT SIXTY THOUSAND BILLION MILES

Continuing my celestial voyage, I left the system of Alpha Centauri to penetrate into the starry depths of the Southern Cross. I traversed sunny shores and deserts of night, passing from sun to sun, from system to system, flying past stars which blinded me one moment and then were engulfed by the infinite night. The normal state of the universe is night and silence. There is no light except round the suns and planets; there is no sound but in their immediate neighbourhood, in their atmosphere. In skirting stellar groups, I noticed enormous globes rolling in a strange light, and I often seemed to feel electric shocks, magnetic disturbances, certain indefinable sensations which warned me, by a sort of malaise, that such spheres are unsuitable for our mode of existence, and that they are inhabited by beings whose perceptions, feelings, and thoughts differ from ours. I remember particularly having seen in the course of my flight a group of many-hued worlds illuminated by three suns, one a ruby red, one an emerald green, and a third a sapphire blue, and so singularly illuminated by this false light—false to us, but natural to them—that I asked myself whether I was not the victim of an illusion and whether such creations really exist, though, indeed, having observed those well-known associations of coloured suns hundred of times in the telescope, I ought not to have been in doubt for an instant. I stopped and approached one of those worlds and saw that it was inhabited by beings who seemed to be woven out of light. To their eyes, certainly, the inhabitants of our planet would appear so sombre, heavy, and coarse that they might legitimately ask whether we were alive and whether we felt ourselves to be alive.

Those are worlds peopled by aerial organisms whose brightness surpasses the tint of the freshest roses and purest lilies. These beings live on the very

atmosphere which they breathe, without being condemned, like the inhabitants of our planet, to be constantly killing innumerable animals with which to fill their bodies.

Their beauty, delicacy, and brightness reminded me by contrast of the conditions imposed by terrestrial life. I remembered that brute force reigns supreme here, that millions of beings are killed every day to assure the existence of the rest, that war is a natural law amongst animals, and that humanity is so little freed from animal barbarism that nearly all people continue to accept, as in primitive times, slavery and servitude. Being so far from the Earth, I judged of the colossal stupidity of the inhabitants of our planet. But if, down to our own times, the nations have made their greatest glory consist of international butcheries, that state is transitory. Every tree bears fruit after its kind. Tortoises and bears cannot aspire to the wings of the swallow or the song of the thrush. The military glories of Alexander, Cæsar, Charlemagne, Napoleon, or Bismarck, being of the order of carnivorous animal instincts, last no longer than the brutal repast itself, and a few centuries suffice to efface them from the history of the planet.

Endeavouring to estimate the real importance of this history and of our planet itself, I searched space, not only for the Earth which had become invisible long ago, but even the Sun; but I could neither find the Sun nor any of its brightest neighbours such as Alpha Centauri or Sirius, nor any of the stars which one sees from the Earth. The whole region of space where our floating island gravitates had disappeared long ago as an insignificant point in the depths of space. Austerlitz, Waterloo, Sevastopol, Magenta, Sadowa, Keichshofen, Sedan, were but microscopic agitations in a Lilliputian ant-heap, amusement of infants delighting in muscular exercises involving blood and smoke. Why blame them? Why pity them? They do what pleases them and nobody forces them to do it. Why should astronomy use a magnifying-glass to study the microbes on a planet? The system of many-coloured suns, the blinding organic wealth of which had inspired me to return to the earthly twilight, revolves at a distance of 60,000 billion miles. Light takes more than ten years to traverse this distance. Yet this is nothing extraordinary in the way of astronomical distance.

Sirius, the most brilliant star of our sky, transported to that distance, would be 3,500 times farther away than it is in reality, and it would send us 12 million times less light. It would be a small point, still within the range of the new photographic processes. It would be a telescopic star of the 18th magnitude.

This sidereal milestone would be far from marking the limit of the space accessible to telescopic investigation, which includes stars of the 20th magnitude, and which, according to ingenious calculations, is occupied by about 100 million suns. And indeed, as I advanced in my celestial voyage, I crossed new abysses and discovered far ahead and above me new stars which became suns, shone in the night and appeared single, double, treble, quadruple, even quintuple, radiating a silver or golden light, or emitting the most vivid and various colours; and I guessed in passing at celestial earths peopled by unknown humanities floating in these rays, before these worlds in turn rolled away and disappeared beneath me in the night. They rushed with different speeds in every direction through space, like luminous globes in the bouquets of fireworks, and seemed to fly away in a starry rain.

When I reached the confines of our sidereal universe, the suns and systems became sparser, and as I continued my ascent I found myself engulfed in a black and desert void whence I could see the outer form of our universe, resembling one of those many star clusters which are seen in every telescopic field. This cluster became smaller and smaller as I flew on into the outer darkness.

Then, in the infinite night I perceived above me another universe which appeared in space as a pale and distant nebula, and I understood that all we can see with our eyes in the clearest night and all that telescopic vision has yet allowed us to discover represent nothing but a local region in an animated immensity, and that there are other universes besides that of which our Sun forms a star.

(7) IN INFINITE SPACE

I approached this second universe, which became larger and larger like an archipelago of stars, and I soon arrived at its outskirts. As I traversed it from end to end I saw that it also was composed of several million suns separated from each other by thousands of millions of miles. Then I found beyond it another dark abyss resembling that which I had crossed to reach the second universe.

Continuing my flight, I saw a third, and I crossed it. A fourth approached, then another, and yet another. And as I crossed those deserts which separated them, in whatever direction my gaze endeavoured to pierce the void, everywhere it discovered new universes in the distance.

The splendid spiral nebulae are not balls of gas but agglomerations of suns, Milky Ways situated outside our sidereal universe.

Then I understood that all the stars which have ever been observed in the sky, the millions of luminous points which constitute the Milky Way, the innumerable celestial bodies, suns of every magnitude and of every degree of brightness, solar systems, planets, and satellites, which by millions and hundreds of millions succeed each other in the void around us, that whatever human tongues have designated by the name of universe, do not in the infinite represent more than an archipelago of celestial islands and not more than a city in the grand total of population, a town of greater or lesser importance.

In this city of the limitless empire, in this town of a land without frontiers, our Sun and its system represent a single point, a single house among millions of other habitations. Is our solar system a palace or a hovel in this great city? Probably a hovel.

And the Earth? The Earth is a room in the solar mansion—a small dwelling, miserably small.

Thus in the general economy of nature our planet has no more importance than a poor little room in a considerable house. That house in turn is lost in the middle of an immense town. And that immense town, which to us represents the entire universe, is in fact nothing but a universe beyond which in every direction there exist other universes.

How far is this reality from human pretensions, both ancient and modern, which imagine that our world represents the infinite, that God stops the Sun to illuminate one of Joshua's battles—a miracle renewed, says history, for Charlemagne and Charles V—and that the great Sower of stars took upon himself a human shape to dwell among us!

What simplicity among sincere theologians! What imposture among the chiefs of states who still dare to invest themselves with titles of divine mandates to enslave the people! Are not the real atheists those either ignorant or insincere people who make the sublimest idea the accomplice of all their mediocrities, and are not the real Deists the independent searchers whose sole ambition is laboriously to look for the causes and gradually to work up to truth?

With what strange religious systems has humanity up to now enveloped its barren imagination! The Israelite who believes he is agreeable to God in

practising circumcision or in buying a new knife to be sure that it has not touched pig's fat; the Christian who imagines he can make God descend upon a table and who is told by his preachers that prayers and fasts have an influence upon the weather and agriculture; the Mahomedan who sees the gate of Mahomet's paradise opening before him as he stabs a missionary; the fanatic who casts himself under the wheels of the Juggernaut; the Buddhist who remains fascinated in the beatific contemplation of his navel, or works a prayer-mill for the remission of sins—these surely form the most ridiculous and infantile ideas of the unknown and unknowable Being.

All these littlenesses are related to the primitive illusion of the smallness of the universe, which was considered as a sort of screen studded with golden nails and enclosing the earth in its centre. Certainly if astronomy had had no other result than to enlarge our general conceptions and show us the relativity of terrestrial things in the bosom of the absolute, to deliver us from this ancient slavery of thought and make us free citizens of the infinite, it would deserve our veneration and our gratitude, for without it we should still be incapable of forming true conceptions.

Some conservatives will perhaps object that there are even in French observatories astronomers who go to Communion, tell their beads, and carry candles in the churches, and that the same mentality can be found in certain English, Spanish, Italian, German, Russian, and other observatories. Yes, without a doubt, the fact is undeniable. Such a psychological phenomenon has two explanations. Either these hybrid beings are sincere or they are not. If they are believers, they are illogical and in perpetual conflict with their scientific reason, and we must not be astonished at the strange arrangement which their conscience is capable of constructing between two conceptions of nature which directly contradict each other. While incapable of explanation, such sincerity can be respected, like that of the innocent infant who believes all one tells him.

In the second, case it is hypocrisy, falsehood, rascality, personal interest; and this sort of conscience is suitably judged by every honest man.

These anomalies and limitations have not hindered astronomy from bringing light and independence to spirits who can understand it and who have the courage and the freedom of their opinions.

But in telling of my Venetian dream I did not want to indulge in irritating polemics, and my only object was to show to some open eyes the horizons of

astronomical philosophy, and I hasten to return to my sidereal voyage and to describe its last phase. I shall, however, add another word concerning our tiny planet, remarking that its inhabitants are as a rule so unintelligent and so incapable of judgment that they imagine that all have an equal intellectual and moral value and that in the most civilised country of this globe the vote of an imbecile or a drunkard has the same weight as that of an educated thinker, so that the legislative chamber entrusted with the fate of the country is an incoherent mass of persons incompetent to deal with any of the questions likely to arise. Half of them are ignorant and are preoccupied with their private concerns. In this system of pretended equality, Judas is the same as Jesus, Philip II equals Marcus Aurelius. Torquemada is the same as St. Vincent de Paul, Fouquier-Tinville the same as Mirabeau, and the wine-merchant at the corner is the equal of Archimedes and Pythagoras. There is no reasoning. Our little planet is as insignificant morally as it is physically. Among terrestrial humanity only one man in a hundred is intelligent. Humanity is practically no older than four or five years as regards intelligence compared with what it ought to be normally.

I had traversed several universes analogous to our galactic system, universes separated from each other by abysses of nothingness, and what had struck me most in this general survey was meeting a number of humanities foreign to our own living in various regions of space, living their own lives and carried along to their destiny in the whirlpool of their personal affairs. While the inhabitants of the Earth reduce creation to their own size, thousands, millions, billions, of other humanities live in every degree of intellectual advancement on solar systems which to them are the very centre of observation and from which our terrestrial home appears lost in the infinite distance.

I also saw dead worlds. It is a fact worthy of attention that all existence tends towards death. Beings only come into existence to die. The worlds only attain periods of vitality to descend again into decadence and the tomb. Suns only burn to be extinguished. Death would therefore be the supreme law, the final result.

The mathematician can now calculate with great accuracy the date when our Sun will become extinct and when the Earth will roll on through the eternal night like an icy cemetery. The entire history of terrestrial humanity will have arrived at an absolute zero. The time will come when even the ruins will be destroyed.

On account of the tendency of energy to establish itself in a state of stable equilibrium in the universe, life will have an end on our planet as well as the other worlds.

If everything appears thus to tend to extinction and death, it is because we do not know the secret of the conservation of energy. An end such as I have indicated is really unthinkable. The terms of the problem contain their own condemnation. It is admitted that force and matter can be neither created nor destroyed, and have therefore existed and acted from all eternity. If therefore the final result of the radiation of suns into space is their extinction and consequently the extinction of life on their attendant planets, then since an eternity has already elapsed during which energy has tended to equilibrium, there ought to be not a single sun, star or planet in existence.

Now relatively, not to an eternal duration but only to a period like a lightning flash compared with that, say a trillion years, the life of a human race, of a planet, or even of a sun, is very short. Geologists talk of 20 to 30 million years for the whole duration of the geological eras from the origin of life on earth; physicists talk of 100 million for the constitution of the terrestrial globe from the liquid to the solid state; astronomers also assign 100 million years to the age of the Sun, and even less to its future duration. Even if we doubled or trebled these numbers, even if we multiplied them by ten or even a hundred, we should not arrive at the millionth part of a trillion years! Thus without going back to a previous eternity, if the energy of suns had no other final result but extinction we should not exist now and nothing that is would be.

The universe was not made of one piece at the origin of things. This origin does not in fact exist. We find in space suns of every age. There are old ones, there are new ones. Here are cradles, yonder are tombs. If the first creations formed by matter and energy had not been renewed, there would no longer be a universe. All primitive energy which had animated the suns would be used up. Besides, matter and energy are but one.

Just as in passing through a forest we find oaks in decay, green trees, and new growths, thus also does the celestial traveller encounter in space worlds long dead, dying worlds, worlds in full activity, and budding stars.

Everything dies, but everything lives again.

Among the last worlds in full vitality which I visited on my voyage among distant universes there was one which appeared particularly remarkable on

account of the high state of social progress. Although this world is the most distant of all those suspected to exist in the depths of space, yet the human race which dwells there is not very different from ours, physically. It is divided into two sexes, and the organic forms somewhat resemble those of our race. But the social state is distinctly superior to ours.

A perpetual harmony reigns among all the members of this vast family. Simple and modest, each of these beings has no higher ambition than gradually to raise himself in the knowledge of things and in moral perfection.

The atmosphere is not entirely nutritious, and there, as here, one is obliged to eat in order to live. But they live exclusively on fruit and vegetables, and kill no living being.

The functions of material life only take up a very short time, and life is mainly intellectual. Instead of personal rivalries, great and small, and the various ambitions which agitate the entire lives of the men and women of our poor little world, those beings are mainly occupied in study and pleasure.

There is no money. There are no rich nor poor. The fruits necessary for nourishment can be picked anywhere beyond all needs. Summer is perpetual and no sort of clothing has been thought of because the bodily forms always keep their beauty, and coquetry would have nothing to conceal.

There is no old age. On reaching a ripe age one goes to sleep and the body dissolves like a cloud, which becomes invisible by the change of state of its molecules.

No law has instituted the marriage bond. As it would be impossible to contract for interest, because there are neither castes nor fortunes, love alone guides the choice. On rare occasions the years reveal some divergence of character sufficient to lead to a desire for another choice, but when this divergence shows itself there is no chain to break. Besides, they always remain lovers and never become married. The desire of change, of variety, of curiosity, hardly arises because the persons who have freely chosen each other love each other beyond all others and have only chosen each other because they knew each other.

Friendships are sure and faithful, and there is no example of treason dictated by the vile sentiment of jealousy.

Contrary to what happens on earth, every person whose life is ruled by the sentiment of personal interest or ambition would be considered as a monster

beyond all explanation and thoroughly despised. In that world they do not, as we do here, meet people who are constantly unhappy on account of a desire to occupy all the best places, are never satisfied with their own lot, and who, being indefatigable opportunists, grab everything in their insatiable egoism, and die full of honours and vanities.

There is no frontier. Humanity forms a single family. Communications are established over the whole globe by a sort of language which passes with the speed of lightning. An administrative council controlled by universal suffrage directs public education, science, art, and justice; and this universal suffrage is enlightened, and exercises its choice among the best and wisest spirits. The dregs of the population are not represented; the deputies do not shine by numbers and incompetence, but by worth. In a country corresponding to France, their number would be reduced from 600 to 100, every deputy possessing a special competence in legislative questions. It is superfluous to add that a Ministry of War has never been thought of there. The people, led by reason, do not follow a fetish. Besides, no patriotic sentiment can there be exploited or brutally debased, since no frontier divides humanity, and patriotic sentiment consists solely in the recognition of intellectual worth.

No institute of so-called official science has been established there. No Sorbonne has condemned the theory of the Earth's movement, no Academy has disapproved of the doctrine of perpetual peace. There are no titles, no decorations. Nothing is appreciated but personal, intellectual, and moral worth.

The word "infallibility" does not exist in the language of that people. Only one religion reigns in their hearts: natural religion, founded upon astronomy. Their faculties, more transcendent than ours, their senses, more numerous and more penetrating, their more powerful instruments of observation, have long ago placed them in communication with neighbouring worlds, and they have been able to utilise astral magnetism for purposes of transport from one world to another.

They have discovered the mystery of the union between force and matter, and know that there is a fundamental identity between them. In their own religions, they have never given God a name, and have never dared to play at a cult, knowing that such puerility and such pride would be unworthy of their merit. Their religion consists in a belief in immortality based on the knowledge of the intimate nature of being, in preparations for the future life, in efforts to make themselves better and more perfect by a continual study of creation, and a mutual love based on an enlightened sentiment of justice and equity.

They consider Reason as the highest prerogative of the human race, and would consider any doctrinaire mad who would forbid the exercise of that faculty for the sake of any religious system whatever.

From that world, nobody has ever yet perceived the Earth, and nobody suspects its existence.

Their senses are more perfect and more subtle. While our human race is less endowed than certain animals which can foresee a storm, the changes of the seasons, and earthquakes, they possess the sense of orientation, which we lack entirely, although on our own planet certain species like the dog, the cat, the pigeon, and the swallow possess it.

They seemed to me absolutely happy, though exceedingly sensitive. They spent the greatest part of their existence amid the most refined pleasures. Their world is a perpetual paradise ever born afresh. Perfumes arise from the bosom of splendid and varied flowers, and woods are balmy with intoxicating spices, and the light of day plays upon fairy-like scenery.

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When I contemplated this marvellous spectacle, I felt surrounded and penetrated, so to speak, with waves of sound which cradled my enraptured soul in the most delicious melody my ears had ever heard. The sensation of a celestial attraction seemed to carry me on a cloud and make me slowly descend towards an island on which a palace of flowers appeared. I felt a sort of electric shock and—I felt myself in a high ogival window of Venice. A gondola filled with musicians was returning from the Lido by the Grand Canal. Groups chanted harmonious choruses, the sky shone with stars, the moon was setting behind the domes, and Mars was descending towards the horizon.

The old clock sounded slowly the twelve strokes of midnight. “Well,” I exclaimed, “I have been sleeping. Here I have been for two hours at this window. Meanwhile the Moon has flown 4,850 miles in its orbit round the Earth, and the Earth has traversed 410,000 miles in its revolution round the Sun, drawn by that wonderful attraction which rules the world across the voids of space; perhaps it also rules our souls through the voids of time. “Thou beautiful starry sky,” I murmured, “who hast taught us so much already, wilt thou not soon solve for us the riddle of the great mystery? Thou art our hope, thou alone canst teach us, thou alone canst open before our eyes the panoramas of infinity and eternity.”

II. THE WORLD OF LONG AGO

CHAPTER II

THE WORLD OF LONG AGO

I HAD a dream, which yet was not a dream.

I found myself as an observer of the world as it was 100 million years ago, inhabiting a planet in attendance upon one of those distant stars of space, in the middle of a sidereal universe analogous to that which exists at present, though it was not the same, for the universe of that time was destroyed long ago and the universe of to-day did not yet exist.

At that time, also, there were stars and constellations, but they were neither the same constellations nor the same stars.

There were suns, moons, inhabited earths, days, nights, seasons, years, countries, beings, impressions, thoughts, facts; but they were not the same as now.

The Earth which we inhabit was not yet in existence. The materials which compose it floated through space in a state of diffused nebulosity, gravitating about the slowly condensing solar focus. There were as yet neither water, nor air, nor earth, nor heavens, nor planets, nor animals, nor any one of those bodies reputed to be simple by chemistry, as oxygen, hydrogen, nitrogen, carbon, lead, copper, etc. The gas (which by its condensations and final transformations was eventually to give rise to the various gaseous, liquid, and solid substances at present constituting our globe and its inhabitants) was a single homogeneous gas, containing within it, like an unconscious chrysalis, the possibilities of the future. But no prophet could have foreseen the unknown which slumbered amid its mysteries.

Our planet showed at that time the aspect of those vague gaseous nebulae which the telescope discovers on the floor of the skies and which the spectroscope analyses. The solar nebula in the course of condensation floated among the stars. All humanity with its history, every one of us with all his energies, all the living beings of this Earth, were contained in the germ of that nebula and its forces; but the beings and the things which we know were not to come into existence until after a long incubation of centuries. In the place of what was destined to be the Earth, there was nothing but a gas floating in

starry space. Nor was it in the real “space” where we are now, for the Earth, the planets, and the whole solar system came from afar and flew swiftly across the void.

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In the history of creation, 100 million years pass like a day; they dwindle and disappear, a fugitive dream, into the bosom of that eternity which absorbs all.

Then, although our planet did not exist, there were stars, suns, solar systems, and inhabited worlds as there are now. The humanities which peopled those worlds lived their lives as we live ours.

It would have been a wonderful spectacle for the thinker to contemplate the great work of all those beings. In passion or indifference, in pleasure or pain, in laughter or tears, in movement or repose, they lived; fighting, forgiving; accusing, forgetting; loving, hating; being born and dying; drawn into the fatal whirlpool; blindly succeeding each other through the generations and centuries, not knowing what gave them birth; ignorant of the fate in store for individuals and souls; playthings of that Nature which forms worlds and beings, stars and atoms, centuries and minutes, like soap-bubbles blown by a child into the air; and all plunging into the sea, like those whirls of sand which the desert wind raises and blows along in the typhoon or the breeze. It is the same spectacle as that which the earth offers to-day; living multitudes fighting for life and knowing only death.

The thought which must strike us most in our retrospective contemplation is that at that time the *Earth did not exist*. Not one of those human beings who live now, who will live in the future, or who have lived in the past, were then thought of. Nothing of all that now exists around us existed then. Yet in those ancient worlds which have disappeared long ago, the humanities which animated them had their vivid history, with flourishing cities, fights and struggles, laws and law-courts, judges of spiritual things, historians, economists, politicians, theologians, literary men, who took pains to tell the true from the false and to write down conscientiously what they, too, called “universal history.” For them, all creation stopped in their era and in their place; for all of them, creation was finished; the rest of the universe and of limitless eternity was lost in insignificance in comparison with what they

called the "Present." They never thought of the eternity which had already passed before them, nor of the eternity which would come after them.

They lived, learned or ignorant, famous or obscure, rich or poor, opulent or miserable, religious or sceptical, they all lived as if their era would never come to an end. Some of them, without losing a minute, amassed a fortune which their heirs hastened to dissipate; some spent their time in dreams and contemplations without thinking of the morrow. In one place there would be battalions inflaming the populace with their patriotic shouts; in another loving couples united their souls in mystery. Under the pressure of what they believed to be affairs of imperative importance, driven by the attractions of pleasure or borne on the wings of ambition, the inhabitants of that ancient world, like those of ours, flung themselves into the whirlpool of life. They, like ourselves, had days of glory and of sorrow; they had their '89 and '93, their Austerlitz and Waterloo, and political drama had its 18th Brumaire and its 2nd of December. Thus recently on our own Earth shone the life of Babylon, of Thebes, of Memphis, of Nineveh, of Carthage, the glory of Semiramis, Sesostris, Solomon, Alexander, Cambyses, and Cæsar; and to-day the silence of funeral solitude reigns supreme over the ruins of the palaces and temples, in the slumber of the invading night. In the history of the universe it is not only peoples, kingdoms, and empires which have disappeared, but it is whole worlds, groups of worlds, archipelagoes of planets, visible universes!

For eternity did not *begin*, it was never begun. The forces of Nature have never been inactive. For Nature itself, our measures of time, our conceptions of duration, do not exist. She has no past and no future, but a perpetual present. She remains immutable throughout her incessant manifestations and transformations. We pass away; She remains.

One can hardly think without terror of the innumerable beings which have lived on the worlds now lost, of all the leading spirits who have thought, acted, guided humanity in the path of progress, light, and liberty. One cannot think of Platos, Pascals, and Newtons of the vanished worlds without asking what has become of them. It is easy to reply that that is nothing, that they died as they were born, that all is dust and returns to dust. It is an easy answer, but it is not satisfying.

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Certainly, nobody can be so foolish as to claim to have found a solution of the great mystery. For treating those profound problems of eternity and infinity, we are about in the position of ants attempting to gain a knowledge of the history of France. In spite of all their mental gifts, which have indeed been fully recognised, in spite of their goodwill, their gallant attempts and all their efforts, it is quite probable that they would not get beyond the history of their ant-heap and would not arrive at any reasonable conclusions concerning human beings and their affairs. To them, naturally, the proprietors of the woods and fields are the ants, and the plant-lice domesticated by them. And the parasites of the Earth are those inedible insects which interfere with them. Do they know that birds exist? It is doubtful. As regards men, they do not know of their existence, though it may be that the ants in civilised countries have in their antennal language an expression corresponding to the idea of “sugar-maker,” or cook, or confectioner, or for some implacable enemy such as a gardener. But even if they suspected our existence, they could not form any idea about the human race or its history but—the ideas of ants.

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It would no doubt be as useless as it is foolish to lose ourselves in the nebosity of metaphysics to attain a solution which will escape us for ever. But it is no doubt a proper subject for the exercise of our mental faculties to think of this particular aspect of creation: *Time*; to think that from all eternity earths inhabited like ours have floated in the light of their suns, that from all eternity there have been humanities enjoying the pleasures of life, and that from all eternity the end of the world has sounded on the hoary timepiece of destiny, burying in turn the universes and their inhabitants in the tomb of annihilation and oblivion. For it is impossible for us to conceive a commencement which was not preceded by an eternity of inaction, and as far as the observational sciences can take us, they show us forces in perpetual activity.

If infinite space dazzles us by its limitless immensity, an eternity without a beginning and without end arises, still more formidable perhaps, before our terrified gaze. The voices of the past speak to us from the abyss: they speak of the future.

The past of extinct worlds is the *future* of the earth.

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In 100 million years, the earth where we live will no longer exist, or if any wreck of it remains, it will only be a funereal desert. The various worlds of our system will have achieved their circle of life, the histories of its human race will long ago have been finished, our own Sun, no doubt, will have lost its light and will roll along a dark star, through the realms of night. It may be that, thrown back by destiny into the melting-pot of perpetual change, united in a supreme climax with some old dead sun traversing the same abyss, it will arise like a phoenix from its ashes by the conversion of motion into heat.

But, then as now, the nebulae will have given birth to suns, then as now, endless space will be filled with stars without number gravitating in the harmony of their mutual attraction, the planets will swing in the rays of their suns, mornings and evenings will follow each other, blue sky will spread overhead, clouds will float in the twilight mysteries, perfumed breezes will blow through the woods and valleys, mysterious sounds will stop the songs of the birds, and eternal love will sway a later youth with the divine rapture of insatiable aspirations. Marvellous ascension of life! Nature will chant, as it does to-day, the hymn of youth and happiness, and an imperishable spring will bloom for ever in this immense universe where the historian of the past sees nothing but tombs!

If there are no limits to space, if, whatever part of the sky our thought may essay, it can always pass on without being stopped by anything, however swift or prolonged its flight, if, in a word, space is infinite in every sense, it is the same with eternity: there is no possible limit to it, and whatever end we may imagine, whatever hour or minute fixed for its end, our thought immediately leaps the obstacle and continues on its way. Infinity even now is filled with budding worlds, worlds reaching maturity, decadent worlds, dead worlds, disseminated in all regions of an unlimited space, gaseous nebulae, hydrogen suns, oxidised stars, planets in the course of formation, congealed satellites, disintegrated comets—the forces of Nature are everywhere active, the energy of creation remains constant, neither increasing nor diminishing, and all the scientists agree in testifying that what we call destruction and annihilation is only transformation. Astronomy reveals to us Time as it has revealed Space. It shows that there is nothing peculiar about our present epoch in the history of Nature, nor about our present position in space, and it combines Time and Space, the two forms of reality, in the same synthesis as the two grand aspects of the development of the universe.

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No, this dream was no dream. For to the human race which lived on the different worlds of space during the ages preceding the formation of our solar system, the Earth with all its history was only a possibility of what the future might bring forth. It might never have existed at all. The writers of history of terrestrial peoples, Moses, Herodotus, Manetho, Ma-Tuan-Lin, Titus Livius, Tacitus, Gregory of Tours, Bossuet, all those who have attempted to write universal histories, the great Leibnitz himself, who placed the commencement of the history of a small German duchy at the time of creation of the world, and even the delightful author of the *Metamorphoses*, who tells of the history of the birth of the Earth and Heaven—the astronomer smiles at their annals, as he has smiled at the genealogies of the kings and the conquests of the Cæsars

“Battles of ants in microscopic space.”

Simple illusions of infants who fondle their dolls!

Let them invent new microscopes for distinguishing Charlemagne and Napoleon in the ant-heap of Lilliput; we cannot find them! And the whole Earth, where is it? By an abstraction of thought, we manage to live before it and after it; its whole history has disappeared like a lightning flash which passes in the calm of a long summer day.

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As I contemplated those panoramas of time and space, as the bygone ages passed slowly across my view, with their long trails of past glories, and as the races which peopled the worlds arose from the depths of space, shedding their winding-sheets and walking again in the flowery paths of life, all this prodigious secular past became present, and the millions of suns extinguished through the ages lighted up and shone again. The sky was bright with innumerable stars which our eyes had never seen, and the light of life shed its rays on celestial shores stretching away to infinity!

Suddenly, an immense black veil fell from the skies and hid the view, and I saw no more. In front of this veil, our planet flew along with its speed of 62 thousand miles an hour.

I found myself again in the ordinary condition of an Earth-dweller, who sees nothing beyond the horizon, and who imagines that, in time as in space, our mediocre humanity exists alone on the world.

III. THE WORLD TO COME

CHAPTER III

THE WORLD TO COME

The future is as real as the past. The world to come is in the mind's eye as substantial as the world of long ago.

ONCE upon a time, in a solar system belonging to the constellation Andromeda, a planet a million times the size of our Earth bore on its surface a very advanced human race. The eyes of its inhabitants were constructed differently from ours and received radiations which are dark to us. Also, instead of five senses those human organisms possessed twelve. Their subtle and far-seeing industry had invented instruments of great space-penetrating power, and they had succeeded in determining at immense distances the volumes, masses, densities, physical and chemical constitutions, the movements, and the intrinsic nature of worlds at present quite indiscernible to us.

Amidst the glories of a sumptuous civilisation, those human beings, whose form did not at all resemble ours, thought, in spite of their progress in astronomy, that they were the centre, the final goal, and the justification for the existence of the universe. Some of their philosophers had indeed put forward the idea of a probability of inhabited worlds, but this idea, received with scepticism by most of the learned men and resolutely rejected by the theologians, was only accepted by the most liberal spirits with a reservation concerning the intellectual superiority of their race, considered as the necessary and normal type of all humanity. To them it seemed impossible that Nature should create anything other or better than what had been established in their own world; the zoology of their own planet set a standard, and living beings, they thought, could not be organised otherwise than as they knew them. To their minds, the area accessible to their observation included all the possible manifestations of the forces acting throughout the cosmos. It was only possible to have twelve senses, neither more nor less.

There came a time when some transcending genius discovered among the stars of the constellation which terrestrial astronomers call Centaurus the star which we call the Sun and around which we gravitate. He noticed round this

star nine principal spheres circulating round it, and, by some secret sympathy, he directed his attraction chiefly to the globe which we now inhabit.

The star—our own Sun—and the nine planets we have just mentioned were invisible in our sense of the word. They only emitted dark rays, black light. For a long time already, in fact, the Sun had been extinct, and the humanities who had lived on the surface of the Earth, of Mars, Venus, Mercury, Jupiter, Saturn, Uranus, and Neptune, of the trans-Neptunian planet and its sisters, had died in turn and had been gradually erased from the great book of universal life.

But by means of the superior methods of investigation which this Andromedic astronomer had at his disposal, he succeeded, after a laborious study which it took him 250 years of continuous work to complete, in reconstructing the history of the terrestrial globe, which interested him specially, and in discovering that it had been formerly inhabited by animals of different species, and in particular by certain bipeds endowed with relative intelligence.

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This obscure globe, a black bullet revolving round another black bullet, had a whole history of its own. It had contained in former times an intense and luxuriant life. The springs and summers had brought forth a profusion of flowers and fruit in the sunlit fields, the land had unfolded its golden carpet of corn, springs had murmured among the hills, birds had sung in the trees, the perfumed breezes of meadows and woods had been wafted through the valleys, rivers had rolled through the vast plains, villages and towns had grown up along their banks, human communities had gradually peopled the world, inventing fruitful industries, delightful arts, brilliant sciences; prodigious cities had, in different ages, raised palaces for kings and temples for gods; Memphis had succeeded Babylon, Athens had followed Memphis, Rome had cast Athens into oblivion, Paris had eclipsed Rome, and had vanished in its turn; hundreds of millions of brains had thought, hundreds of millions of hearts had beaten, eternal loves had been sworn, divine embraces had united loving souls, innocent children, charged by the light of day, had held out their arms for the kisses of their mothers, and life in all its forms had sparkled for millions of years in sheaves of light ever renewed like universal and inextinguishable fireworks. Struggles, miseries, lies, rivalries, ambitions, battles, despairs, tears,

mournings, had too often disfigured with black tempests the sky whose clearness had, in the springtime of life, seemed unchangeable.

What had dominated everything was a wise and impenetrable trickery by which Nature persuaded all young girls to become women, to adorn themselves with irresistible allurements, and to open their arms to men in order to assure the continuity of life, hiding from their truthful eyes the dangers and sufferings to which she condemns them by surrounding them with flowers. And thus humanity had continued without a stop, believing that its destiny was to enjoy without end and progress without a limit; and thus it had finally reached the annihilation of the race and the planet, without leaving anything behind of its splendours and its conquests. What is past is past. Neither terrestrial humanity nor its abode remained. All had disappeared, all had been suppressed, except the spirit. The universal spirit still reigned. But the metamorphoses of matter had transformed everything. The entire history of our globe had been wiped off the slate by the sponge of Time; and the sidereal universe went on as if that history had never been written.

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For several millions of centuries, the Sun had been a globe of gas, shedding light and heat around it. After being a brilliant white it had become yellow and then red, passing, in the course of its cooling process, through the successive stages of white, yellow, orange, and red suns like Sirius, Arcturus, Betelgeuse, and Antares. As it had grown cooler, terrestrial life had become attenuated. The Sun had finally been covered with a solid crust, often pierced by the pressure of the incandescent lava within, and giving rise to prodigious volcanoes. With the failure of light and heat, the joys and pains of terrestrial life had come to an end. And the radiant day-star of former days had become an obscure globe covered with oceans and clouds, without a new sun to illuminate it, without day or twilight, careering through space in the eternal night, and gradually enveloped in a winding sheet of ice and snow consisting of carbonic acid. All the nations of the Earth had gone to rest in as many cemeteries. The dying Sun had recapitulated in its evolution the phases of its ancestors. In infinite space, the extinct suns are much more numerous than the luminous suns, and the stars revived by collision with another are the exception. Temporary stars, which only shine a short time, are exceptional occurrences.

Thus our extinct Sun still roamed through the void, carrying along its retinue of defunct planets and travelling with great speed through empty and unconcerned space.

And at the time of which we are speaking, the stars still shone in the sky, the worlds still gravitated around the suns in space; but they were no longer the same stars, nor the same suns, planets, or humanities; it was neither the Earth nor its contemporaries. Life continued to blossom; but it was not our life.

Just as, before the birth of the Earth, other worlds had flourished in space, so also after the death of our planet will the universe continue to exist, as it existed during the human era. And in the world of which we speak, a new and flourishing humanity shone in the joy of another sun. What had happened was in direct opposition to what terrestrial theologians had taught us concerning the end of the world. For them the end of the world was to have meant the end of the living universe and the establishment of a celestial and infernal world. For every one of the mortals inhabiting the future globe of which we speak, life passed with the fugitive and inexorable speed of the river which flows from its source to the sea, day by day, month by month, year by year, so swiftly that at the end of its course all the moments of that life seem to touch.

The inhabitants of the world of Andromeda lived on their immense globe and occupied themselves with their personal affairs as if our Earth had never existed and without suspecting that long before them, in the past, our human race had played the game of destiny, cradled in the illusion that they existed alone in the world. Nobody thought, amid the common people, that worlds succeeded each other in time as well as in space. Only the thinkers standing on the heights from which the whole of things and events can be surveyed, realised that important truth, that the doctrine of the plurality of worlds applies to eternity as well as to infinity.

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In our attempts to arrive at an idea of the constitution of the universe, two questions constantly and inevitably present themselves to our minds: the questions of Space and Time. They are correlative, but their interpretations are far from being identical, as is often supposed.

Space does not exist by itself, nor does time.

It is impossible to imagine the suppression of space. It is asserted in vain that space is the interval which separates two objects, and that if the objects are

suppressed, space vanishes also. That is a pure scholastic sophism. The definition is not exact. One can safely say that it is impossible to suppress by a thought the place where objects could be. The place, the locality, that void, if you like, by whatever noun it is designated, is there quite ready to receive any object which our imagination can suppose to be there. Even though the universe did not exist, if nothing, absolutely nothing, existed, that nothing would still be space, empty space ready to receive an object. We are then forced to conclude that space exists by itself, even if it cannot be measured in any way.

It is not the same with time.

Time is created by the movements of the heavenly bodies. If the earth did not turn, nor the stars, if there were no succession of periods, time would not exist. It is astronomy which has created time.

If you suppress the universe, space continues to exist, but time ceases, vanishes, disappears.

We measure the duration of a second, of an hour, of a day, of a year, because the heavenly bodies are there as points of reference between which we can count. Besides, it is all relative. If the Earth moved twice as slowly, the days would be twice as long, but would be apparently the same. If our calendar were different, it would still be our measure of time, and we should be none the wiser.

One may put all the clocks forward or backward by one hour, and all the calendars by one day, one month, or one year, change the reckoning of the centuries, or modify the time system in any way one pleases, but it would not change the real course of nature.

The terrestrial days and years do not count in the heavens. The time which can be measured in our own system on the nearest planets, Mars, Jupiter, Saturn, Venus, Neptune, although simultaneous with that which we measure here, is not relative to ours. Time is essentially local. If we did not exist there might be other measures of time, but it would not be time according to our conception. In fact our impressions are relative, there is nothing absolute in time. If we suppress life, the sensation of time disappears, time itself ceases to exist. In empty space, a thousand centuries are no longer than one minute, because they do not exist.

It is we who say “yesterday” and “to-morrow.” To nature, everything is “to-day.” Besides, every one of us has had the opportunity of proving for himself that time does not exist when we sleep. To sleep one hour or five hours is the same to us as regards the appreciation of time. Time being purely relative, a sleep of a decillion years would be the same as the sleep of an hour.

Renan the philosopher, who expressed this truth, added: “Heaven does not exist; in a decillion years it may possibly exist. Those whom a tardy justice places there will believe that they have died the previous evening. To have been, means to be. Succession is the absolute condition of our mind; but in a material object, succession and simultaneity are confounded. When in the presence of death, we ask whether the night will be long, we are as simple as the child who asks the same question in going to bed, because he loves the daylight in which he plays.”

If our thinking monads are associated with these worlds to come, eternity is their empire.

Speaking in the absolute, time does not exist. But space exists.

It might be objected that space itself is a measure, and when we cannot measure it we cannot know it. No doubt if the terrestrial globe were 100 times smaller, 8 feet would barely be as long as one inch, and the man measuring “6 feet” from head to foot would really only be $\frac{3}{4}$ inch high. But nothing would appear smaller, because the metre would still be the ten-millionth part of the quarter circumference of the Earth, and everything would be reduced in the same proportion. As in the case of time, measurement is essentially relative and has nothing absolute about it. But this does not alter the fact that time only exists by the succession of events, whereas space exists absolutely.

Empty space, which is nothing to our senses unless we measure it by some length, cannot be suppressed. Whether we measure it or not this is nothing to do with its existence by itself. These measures of space which we take have no common measure with infinity, nor with absolute space; yet they are taken in absolute space, and depend upon our means of observation. In the infinite void we can imagine several measures of space, all very different: for example, a fourth dimension which for us has no dimensions at all, and in which modes of investigation unknown to us can discover other dimensions which we cannot even guess at, since for us three dimensions exhaust all possible measurements of space.

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The past which is no longer in existence, the future which is not yet, is contained as a germ in the present. To the universal eye, everything is in the present. In transporting ourselves, as we have done, into future times, we observe the events of those future times as if they were already present and already past. We are ephemeral atoms floating in the bosom of eternity, seen for an instant in a beam of light. We regard our epoch as a permanent reality—the illusion of a grain of dust which appears and disappears in the beam.

He who contemplates nature must live in those ages as yet uncreated as well as in those which have passed away. And the future as well as the past are even more real than the present, which does not exist, since from one second to another time climbs up into the future only to fall Lack into the pit of the past. Shall we say that the “present” is the present hour? No, for an hour is long. The present minute? No, for the minute is long to the observing astronomer or physicist. The present second! No, for it is exceedingly long to electrify. Shall we reduce the “present” to the tenth of a second? Yes, if you like, but it is still relative to our sensations. Still, let us agree to that. Here, then, is the present—a tenth of a second! All the rest is past or future, and eternity is the only permanent reality.

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Thus, from that point of space and time where we were placed in that celestial region of the constellation of Andromeda, we have seen arise before us, like a mummy awakening from a long sleep, the history of the dead Earth, while we saw life shaping itself in a world which, in our twentieth century, was not yet in existence. Thence, reascending through the ages, we watched the slow secular evolution of our humanity, until sunlit days became night and we lived in a new present which, in its turn, seemed unique and eternal.

The world to come is there, in the future eternity, just as the world of long ago is there, in the eternity of the past, and all is present in the Absolute.

God does not look either forward or backward. He does not remember, he does not foresee: he sees. We can write our verbs in the different times, past, present, and future. In transporting ourselves by means of thought beyond even the distant era of which we have sketched an episode, we could describe even this episode as an ancient event.

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The history which we have written took place—a hundred million years after to-day!

IV. VENUS THE BEAUTIFUL

CHAPTER IV

VENUS THE BEAUTIFUL

Εσπερος ος κάλλιστος ἐν οὐρανῷ ἱσταται ἄστηρ.
(Hesperus, most beautiful of stars in the sky.)

HOMER, *Iliad*.

IT was Homer who, 3,000 years ago, saluted Venus as the most beautiful of stars. Who has not been struck with her wonderful brilliance? Who can refrain, when she shines so marvellously in the heavens, from greeting her as the brightest of the stars and asking what mysteries are hidden in that light?

This radiant star of eve has been the first to be noticed since the earliest ages; it is the only planet mentioned by Homer; Isaiah celebrated her splendour under the name of Lucifer; at the time of the pyramids the Egyptians called her “the celestial bird of morn and eventide;” thirty-five centuries ago the Babylonians observed one of its transits across the sun; the Indians called her “the brilliant,” and the Arabs “Zorah, the splendour of the sky.” From the earliest days of the world she was the goddess of beauty and love. Let us raise our eyes to the heavens to-night: there is the star chanted by Homer and Virgil.

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How many events have happened since those far-off days! Nations, languages, religions, all have changed. Where are the eyes which looked upon Venus 3,000 years ago? Where are the hearts which confided to her their vows of love for all eternity? And who will be *our* successors when, 3,000 years hence, the Parisians of the fiftieth century admire, as we do now, the star of the *Iliad* twinkling in their sky? The history of man passes quickly, the waves succeed each other and disappear in the ocean of the ages; the heavens remain, and the astronomer smiles at great ambition and puny achievements.

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Venus passes every eighth year through the period of greatest brilliance (1889-1897-1905-1913-1921-1929). She is then so bright that she casts a shadow like a small moon. This is easily seen either in a dark room or when walking past a wall in the country. She can be seen in daylight with the naked eye, not only before sunset, but at midday if one knows where she is. No star or planet attains anything comparable to such brightness.

This visibility of Venus in daylight has been noticed for a long time; sometimes it becomes a public event, as in the spring of 1905. That year among others our beautiful neighbour was under exceptionally favourable conditions of observation. Everyone could see the radiant planet flaming in the west in the spring; in February, March, and April, the proximity of Jupiter, and sometimes of the Moon showed to all eyes a most charming spectacle. The astronomical ignorance of the inhabitants of the Earth is so universal that a free rein was given to fancy, and in France one could read in the papers, under the title: "The Luminous Phenomenon of Cherbourg," a series of the oddest and most contradictory descriptions. They spoke of an oval disc describing curves in the sky; the appearance of an electric meteor; a halo due to the deviation of the sun; of an illuminated captive balloon, of a new kind of maritime signals, of an unknown star, of a comet, even of a "constellation"!

And there was more to come. On the eleventh day of observation, April 11 (the strange apparition had commenced on April 1, and mariners might have thought of an April hoax), the maritime prefect of Cherbourg ordered the commander of the *Chasseloup-Laubat* to study the luminous phenomenon, A vessel was sent to look for Venus! The naval officers could not explain the mystery; one of them, however, wrote that it might be the planet Jupiter!

Other commanders, having heard of the comet discovered at the Nice Observatory by M. Giacobini, announced that the "unexplained light" might well be that comet! They did not know that that comet was a telescopic one, invisible to the naked eye.

In the night of April 10-11 a meteorite was seen at Tunis. The question arose whether it was not this meteorite which had first been seen every evening at Cherbourg!

The phenomenon was signalled from Perpignan, Montauban, Nantes, le Havre, La Réole, Amélie-les-Bains, etc.

And so on. Every sort of stupidity was given out on the subject.

Well, a star resplendent with light shone every evening in the western sky. It was Venus, the famous Shepherds' Star. It was seen from every point of France, from Europe, Asia, the United States—and from Cherbourg as well. For three months it reigned on high every evening. It was also at its maximum brilliancy, and so bright that it cast a shadow, as we have said. And nobody at Cherbourg spoke of Venus, nobody compared with Venus the new star situated in the same region of the sky, nobody thought that this mysterious heavenly body might be none other than the radiant planet. Nobody seemed to know that Venus was there!^[1]

The story repeated itself from December 1912 to March 1913. Venus queened it over the first hours of the night, moving ever nearer to us and remaining a little longer every evening above the horizon.

On the very rare date recorded by the postmarks as 12-12-12 (12 Dec., 1912), the conjunction of Venus and the Moon attracted general attention, especially as the weather was fine. It was the same at the other conjunctions, January 1912, February 11, and March 11.

This association of radiant Venus with the lunar crescent in the evening sky offers to the eyes the most inspiring of spectacles.

The sensation caused at Cherbourg in the spring of 1905 by the brilliant star of the evening shining upon the sea and taken for a mysterious instrument of espionage was renewed in 1913, especially in England. Those who saw it, blinded by its radiant beauty, were led to confound the Shepherds' Star with German airships, and accused them of espionage.

One could read, in fact, in the paper, that the authorities on the other side of the Channel were greatly alarmed by the nocturnal flight of mysterious dirigibles which came under cover of the shades of the evening and hovered over British ports. The cars and the balloons themselves were, so it was said, invisible on account of the darkness, but the powerful searchlights thrown upon the earth revealed the unusual presence of the aerial intruder!

At the same time we received from Russia similar stories of the fear inspired by Venus among the people, who professed to recognise in its bright light the fires of Austrian aeroplanes spying out the upper atmosphere.

In Roumania, Venus was taken for a Russian aeroplane; and in Bessarabia (Russia), rifle-shots were fired at the beautiful planet, in the belief that it was a Roumanian dirigible.

Yet it would be sufficient to use quite a small telescope or even a good opera-glass to correct this error, for Venus showed phases which would have immediately completed the identification, and certainly not allowed it to be confused with aerial vehicles.

Such popular emotions caused by the appearance of Venus are not confined to our own times. In December 1797 the young General Bonaparte, after his wonderful conquest of Italy, returned to Paris to receive from the Directory the honours which presaged the consulate. Attended by a brilliant staff, he was going on horseback to the palace of the Luxembourg, where the Directory awaited him, when he was surprised to see, in the Rue de Tournon, all the people who stood ready to greet him turn round and look at a point in the sky instead of looking at him. His aide-de-camp told him that a star shone in the sky and that the French saw in it the star of the conqueror of Italy. It was Venus, then at its maximum brightness. Political conditions have changed, but the star remains.

It was there in mythological times:

Venus Astarte, daughter of the sea,
Shook off her mother's tears amid her fond caresses
And fertilised the earth in wringing out her tresses.

She was there, receiving the lovers' holocausts within the temples consecrated to her in Greece, Egypt, India, all over the world, for it was from her brilliant or mysterious aspects of the morning and evening that the cult of her charming personality was derived. Even nowadays many an observer sees in her only a radiant beauty in an ethereal dwelling-place, and does not remember that science has explained the idol and transfigured the star.

From its greatest easterly elongation to its greatest westerly elongation the brightness of the planet is such that when it is not quite close to the sun it is usually seen with the naked eye in daylight. Its phases are always curious and interesting. Their discovery by Galileo was of great importance to the beginning of modern astronomy in proving that the planets are globes without light of their own, similar to the globe which we inhabit.

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Venus is no longer for us an allegorical symbol lost in the incense of the clouds and reigning over enslaved hearts; the Earth is no longer an inferior abode controlled by celestial influence; the horizon is grown wider, our planet has been liberated in unlimited space; Venus has become a celestial Earth, our sister and neighbour, and the better-informed eyes which contemplate it to-day see in her, not, like Homer and Manilius, a luminous point shining above our heads and controlling the feelings of our hearts and the movement of our blood, but a world corresponding to the world we ourselves inhabit, gravitating like us round the same sun, living on the same light and heat, and fit like ours to bear a thinking race for whom the Earth we inhabit is itself a star in the sky.

Venus, the brightest star which ever shines in the limpid glory of the western sky, is not a star, strictly speaking; it is a planet, a world like ours, and of the same size, which only shines by reflecting the Sun's light into space. When one remembers that it is the same with us, and that, seen from the distance of some ten million miles, our Earth shines with a similar lustre, one is forced to admit that we are much more beautiful from afar than we are close by.

There is indeed no possible comparison between the two aspects. Seen close at hand, we are agitation, the struggle for existence, fight, battle, envy, jealousy, drama, hunger, often misery; seen from afar, we are calm, serenity, pure nobility, celestial light, almost an image of God! It is probably the same with Venus, so white and so radiant seen from here; possibly if we could go close to her, we should hear the cries of wild beasts in the forests, the battles of men devouring each other in so-called civilised lands, and we might witness geological and human revolutions, more formidable on account of the fact that Venus, younger than ourselves, is less advanced in evolution.

Being nearer the Sun than we and enveloped in a very dense atmosphere, Venus must have a higher temperature than the Earth. Its atmosphere is heavily charged with hot vapours. Its sky is always overcast; thunder and lightning must be never-ending. Electricity plays no doubt a prominent part. But, although many things are imperfect, there is good everywhere.

It is one of the charms of astronomy that it enables us to see through time as well as through space. Those who remain in ignorance of the elements of this science do not even know that they are depriving themselves of the most agreeable satisfactions of the mind. They are like travellers who pass through a wonderful landscape without even asking where they are. This planet consecrated to Venus, the goddess of beauty and love, who, in the days of

ancient Greece, was said to have emerged from the waves to charm gods and men, and whose mythical history brings us such eloquent evidence of the influence of celestial aspects upon the origin of religions—this planet, we say, owes its terrestrial glory to its situation between the Earth and the Sun. While our globe gravitates round the Sun at the distance of 92 million miles, in a year of 365 days, Venus passes along the orbit contained within our own, at the distance of 65 million miles from the Sun, in a year numbering 225 days.

It follows that from time to time it passes between the Sun and ourselves, approaching to within 25 million miles or even less, since the two orbits are not circular, but elliptical.

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One of the finest achievements of astronomy is having made man a citizen of the heavens. The planet which we inhabit is a heavenly body, just as is Venus, Mars, or Jupiter, and far from occupying the centre of creation, it lies in the depths of infinite space as do the most distant stars of the Milky Way. Venus possesses no more light of her own than does the Earth; she simply receives the rays of the Sun and sheds them into space as the Moon does. Take, for instance, a small finder telescope and direct it towards Venus; you will see the form of a crescent. It is no longer Venus, but Diana. Take a rather more powerful telescope, and you will see that the border of this crescent is not regular, and the southern pole is blunted and rounded, while the northern pole is pointed. On increasing the power of your instrument you will perceive the atmosphere by the gradual transition between the illuminated hemisphere and the dark hemisphere, or by the blinding clouds and light shadows which fleck its disc.

If you go still farther and give yourself the pains and the pleasure of doing a few astronomical calculations, you will see that the diameter of its globe is just the same as that of the Earth—within one thousandth part!—but that Venus is a little lighter than the Earth, its density being only four-fifths of ours, whence it results that objects weigh a little less on its surface than they would on ours: 1,000 grammes transported on Venus would only weigh 880 grammes; on Mars they would weigh even less, viz. 376 grammes; and even less on the Moon, viz. 165 grammes. We are very heavy down here.

Astronomy alone teaches us that this young sister of ours is in communication with us, not only by means of light, but also by attraction, and

that space, so far from being a separation between worlds, is a real link, an invisible hyphen. For example, when the distance between us and Venus is 25 million miles, light only takes $2\frac{1}{2}$ minutes to cross that distance. Is that a serious separation? Even in France a telegram, in spite of its name, takes more than an hour to be delivered a few hundred miles away, on account of intermediate links. If astronomers ever succeeded in establishing celestial telegraphy, communication would be much more rapid between one world and another than between one part of modern Babylon and another. An interplanetary telephone would be almost instantaneous.

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If the luminous beams from Venus to the Earth only take two or three minutes to reach us, according to its proximity, the attraction between the two planets is transmitted still more rapidly, for it does not even take a whole second. And just think! At the distance of millions of miles we feel the mysterious influence of Venus, we abandon for a moment the regular course of our orbit round the Sun to follow in her path. Celestial mechanics calculates this displacement and accounts for it in its determinations.

If the Sun were not the strongest, the two sister-planets, Earth and Venus, would gradually approach each other and would graze one another like two dragon-flies roaming together through the fields of the sky. But fortunately for us, the omnipotent Sun soon resumes its rights, and everything is restored back to order.

The years of Venus are shorter than ours. Their precise duration is 224 days 16 hours 49 minutes 8 seconds. We do not yet know the diurnal rotation, i.e. the length of its days. According to certain observations, this appears to be about 24 hours, while others indicate that this planet does not turn upon its axis at all, or rather, that its rotation is equal to its revolution and that it always turns the same side towards the Sun.

It is extremely difficult to determine its rotation, for two reasons. Firstly, because its atmosphere is very dense and always full of clouds, so that it is impossible to make out its geography and to follow in the telescope the displacement of its surface details; this is so easily done in the case of Mars, whose atmosphere is always clear, and whose rotation is known with perfect accuracy, its period being 24 hours 37 minutes $22\frac{1}{2}$ seconds. The second reason is the difficulty presented by the phases of Venus. The closer the planet

approaches us in its orbit round the Sun, the more does its disc apparently grow in size and the less we see of its surface, because it at the same time passes between the Sun and us, so that its illuminated hemisphere, being naturally turned towards the Sun which illuminates it, is hidden from us.

My personal observations lead me to believe that the period of rotation is about 24 hours. At the Juvisy observatory we have often observed and photographed snowy patches in the north and south, probably marking the extremes of an axis and therefore a rotation.

This special point is not yet decided. It is quite possible that the days are very long there, or rather, that they last for ever—perpetual day on one side and perpetual night on the other side. That would make it a very singular world. On one side light, heat, and life; on the other, the icy coldness of death. Some might choose for their abode perpetual sunlight; others might prefer a night illuminated perhaps by electric light; yet others might prefer to dwell in the dawn or the twilight. Beautiful Venus would then have one hemisphere in perpetual night and one in perpetual day. How strange! How our worlds must differ in the form of their organisms and the nature of their inhabitants!

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The science of the stars opens unexpected outlooks on extra-terrestrial life. We know already that Venus, the Earth, and Mars are three floating homes controlled by the same forces, governed by the same attraction and cradled in the fluctuation of the same magnetism. Regarded from this point of view, Venus is more interesting to us than she ever was in mythological times. Is not the knowledge of those celestial sympathies a first stage of the road leading to the conquests of other worlds?

FOOTNOTES:

[1] The astronomical annuals had announced the brilliant appearance of Venus. But who reads the *Nautical Almanac*? People prefer to read the almanacs which falsely predict rain and fine weather and guess at the political events of the year. Humanity prefers its ignorance and its illusions.

V. THE PLANET MARS

CHAPTER V

THE PLANET MARS

THE inhabitants of the Earth are at last beginning to take some interest in the sky. Tired of living as blind strangers to their own country, they are beginning to know that the world on which they move about is a planet gravitating round the Sun, and that other sister-planets swing round at the same time in the harmony of the solar system. Mars is now spoken of in public as one speaks of politics and of socialism. In America as well as in Europe, at Buenos Ayres, Mexico, or Caracas, as well as at Paris, Milan, Petrograd, Budapest, and Stockholm, the latest telescopic results are discussed. It is known that this neighbouring planet sometimes approaches the Earth within $37\frac{1}{2}$ or even 35 million miles; that astronomers have their eyes on the planet and that they have observed luminous flashes on it, the explanation of which is puzzling them. People remember that in a certain year, 1877, the planet being in its greatest proximity, straight lines resembling canals were discovered, and that the question of possible inhabitants of this new country and future communication with them has been raised. Questions are put and answers are given, discussions give rise to curious confusions and exaggerations, but the net result is that an interest is created in these high questions which raise us above the vulgarities of ordinary life, and a general knowledge of the universe is advanced. That is the main thing.

The last occasions on which Mars was in close proximity to us were 1897, 1899, and 1909.

This remarkable development of public curiosity is easily explained by the marvellous achievements of contemporary astronomy and the admirable precision of certain results obtained. Unless one has a stone instead of a heart and a lump of fat in the place of a brain, it is difficult not to feel some emotion over the achievements of science. If we declare, for instance, that we know the general geography of Mars better than we do that of our own planet, the listener or reader is at first inclined to be somewhat sceptical. But if we show him, either in a telescope or in a diagram, the snows of the north or south poles of Mars, he will admit that nobody has as yet had a complete view of these regions on the Earth, in spite of the discoveries of polar explorers, and he will

be convinced that we do know those Martian regions better than our own poles. That is already a fact of some interest; but we can go farther than that.

It is not only the pole, but the whole surrounding country that is better known on Mars than on the Earth, not only from the geographical, but also from the meteorological point of view. Thus, for instance, we can almost constantly measure the extent of the polar snows, and we find that it varies with the seasons. We see with our own eyes the melting of these snows taking place very rapidly under the light and heat of the Sun, night after night, so to speak, during a summer which is twice as long as ours. The snows disappear almost entirely, and only a little ice remains on a region which we know, and which represents the pole of extreme cold, situated 212 miles from the geographical pole. In spite of the perseverance and heroism of arctic explorers, none of these climatological facts have been witnessed on Earth. It is possible that the Martians are ignorant of their own phenomena if they have been unable to reach their poles. Still, since their poles are free at the end of the summer, they are much better able than we to explore their polar regions. We may say that in general the meteorology and climatology of Mars are better known than those of the Earth. At the time when you read these lines you do not know, and nobody can tell you, what sort of weather you will have to-morrow. But we do know almost certainly what the weather will be to-morrow, next week, next month, on such and such a region on Mars: if we do not wait till the winter comes, we know that it will be fine. Hardly do we see a cloud between the vernal equinox and the autumnal equinox, neither in the equatorial nor the temperate regions, and hardly even in the circumpolar regions. If we are unable to make a drawing of the telescopic image of the planet, the difficulty hardly ever arises from the Martian atmosphere, but from our own, which is so often overcast or turbid. All the geographical configurations, seas, rivers, plains, covered with vegetation according to the moisture available, water-courses varying with the seasons, canals and oases, are mapped out with precision; we know in advance which country will pass across the field of our telescope; and the period of rotation, as already mentioned, is known to the 100th of a second. It is 24 hours 37 minutes 22.58 seconds. We also know that the Martian year contains 59,355,041 seconds, i.e. 686 terrestrial days 23 hours 30 minutes 41 seconds. But since that planet turns on its axis a little more slowly than the Earth, there are only 668 Martian days in the Martian year. In fact the Martian calendar is composed of two successive years of 668 days and a leap year of 669 days. As in our case, there is no exact number of days in the Martian year. Perhaps their calendar has also been reformed several times

without being made perfect. But let us hope that they are not as stupid as we, with our months of 28, 29, 30, and 31 days, and calling the ninth, tenth, eleventh, and twelfth month the seventh, eighth, ninth, and tenth respectively; with our disagreement about dates, Russia only arriving at the 1st of January when the rest of the civilised world has reached the 14th; with our three kinds of days: the civil day which commences at midnight, the astronomical day which commences the next midday, and the naval day which commences the previous midday; we who waited thousands of years before we could fix an exact hour, because we counted from conventional meridians, and the various countries could not agree upon the single meridian. Being probably more advanced than ourselves in its planetary age, Martian humanity is most likely more reasonable and is not mixed up with the littleness of frontiers, dialects, customs, national rivalries, etc. For a long time already, no doubt they form a simple unit. One may also suppose that they do not celebrate their new-year festival and its rejoicings amid the winter frosts, but in the hopeful days of the equinox.

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One of the most curious observations which have been made on this neighbouring planet, or rather which have, apart from the canals, attracted the greatest attention, is that of the luminous flashes. It has been said that these flashes are all seen at the edge of the disc, or beyond it. This is not correct; they show themselves on the line which separates the hemisphere illuminated by the Sun from the dark hemisphere—the line called the “terminator.” They are only seen when the globe of Mars offers a sensible phase, and only along the line of that terminator.

The phenomenon is a slight projection, swelling, or puffing-up of the terminator. It is not a more extraordinary observation than that of the irregularities in the lunar diameter at certain phases: the Sun illuminates, either before its rising or before its setting, the summits of mountains whose bases are still in darkness, and such summits sometimes appear on the Moon as luminous points detached from the disc. Some fertile imaginations have interpreted these flashes as forests on fire or as signals sent out by the Martians. This is going too far. But the possibility of the population of Mars by a human species more intelligent than ours is quite a natural conclusion from the observations. One may also guess without scientific heresy that the canals of Mars are rivers straightened with a deliberate intention of distributing water

which has become a rarity over that planet. The astronomers who deny these possibilities show a very poor spirit. But, on the other hand, there is no reason to see nothing on that world but human activity. Among several explanations of observed phenomena one must always prefer the simplest. In the case of luminous dashes on the terminator, the illumination of mountain-tops or clouds by the Sun suffices to account for them.

Doubts were raised concerning this explanation by the height of 200,000 feet found by an astronomer for the elevation of these mountains. I went over the calculation and found only 15,000 feet. These mountains would not therefore be higher than Mont Blanc, and perhaps less. We should also remember that those luminous projections appear every time that the planet returns to the same condition of illumination with regard to the Earth. They were observed in 1890, 1892, 1894, 1899, 1907, 1909, 1911, 1913, etc. The regions where they appear are a sort of island called Noachis, another called Hesperia, and a third called Tempe. According to all appearances, we have to do with high mountains covered with snow and with still higher clouds.

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The epoch at which the inhabitants of Mars can communicate with us has not yet arrived, or perhaps it has already passed away. All cosmological studies agree in presenting this planet as older than ours, since it is farther from the Sun, and as having passed through its phases of astral life more rapidly than we, on account of its smallness and lightness. We cannot pretend to know the forms which living beings may have assumed there; but we cannot imagine, on the other hand, that the forces of nature, being the same as here and exercised under almost identical conditions (atmosphere, climate, seasons, water, vapours, etc.), have been sterilised by a perpetual miracle of annihilation while on Earth the cup of life overflows all round and the generative force of living beings everywhere surpasses continual and permanent production. But whatever may be the forms of Martian humanity, they must be superior to us, for several reasons. The first of these is that it would be difficult for a human species to be less intelligent than ours, because we do not know how to behave and three-quarters of our resources are employed for feeding soldiers.

The second reason is that progress is an absolute law which nothing can resist. If therefore the inhabitants of Mars have passed their infancy, the centuries have brought them to an age of reason, and their present state represents what our race will be in several million years.

A third circumstance is that they are better situated than ourselves for escaping from the heaviness of matter. A given bulk of water, earth, or other substance is only seven-tenths of the weight it is here; 1,000 grammes taken to Mars would only weigh 376 grammes there; and the woman weighing 8 stone would only weigh 3 stone there.

And, finally, the climatic conditions appear to be much more agreeable there.

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Those are all advantages in favour of the Martians. If, therefore, the idea has occurred to them to make signals to us, it is probably not at the present time. There is no reason that they should think of it at the same time as we and should wait for us. Perhaps they tried 200,000 or 300,000 years ago, before the appearance of man, at the time of the cave-bear or the mammoth. Perhaps they addressed themselves to our planet at the time of the Iguanodon and the Dinosaur. Perhaps they tried again 2,000 or 3,000 years ago. Never having seen any sign of life, they will have concluded that either there are no inhabitants on the Earth, or that they are busy with other things besides the study of the universe and eternal truths. That was true yesterday—and it is true to-day.

VI. THE GIANT WORLD OF JUPITER

CHAPTER VI

THE GIANT WORLD OF JUPITER

EVERYBODY notices, year by year, in a southerly direction, a bright star which half the spectators believe to be the evening star, but which so far from being Venus, is its very antithesis.

To mistake Jupiter for Venus is a sufficiently gross error. But even this error is preferable to nothing. It is better to be mistaken than not to see anything. A large portion of the human race sees nothing at all, thinks nothing, and passes its whole life in the stupidity of plants and slugs.

To notice the star in the sky, even to give it a wrong name, is something, at all events. It shows that one does not go about with eyes cast on the ground or occupied with worldly affairs and with the spirit absorbed by material interests.

Venus, the Shepherds' Star, the morning and evening star, is never far from the sun. It is never in the south, neither in the evening nor at midnight. Its orbit round the Sun lies within our own. Jupiter, on the contrary, revolves round the same Sun along an orbit outside our own, five times as far as the central luminary, at 485 million miles instead of 92 million miles. The distance between us and Jupiter is therefore always greater than 393 million miles except when the Earth reaches the extremity of its elliptical orbit ("aphelion") at a time when Jupiter is in "opposition" to the Sun. We often see it at a medium distance of 450 million miles. If the atmosphere extended over all that distance, an aeroplane flying without a stop at 62 miles an hour, and therefore covering 1,500 miles a day, would take no less than 300,000 days or 812 years to complete the voyage. It would be wise to take some provisions with us!

But on what sort of a world should we arrive?

A giant world, an immense world, a strange world. It is only a thousand times smaller than the Sun—that is to say, 1,000 times larger than the Earth, or even more. Jupiter is eleven times larger in diameter than our globe, i.e. 1,300 times larger in volume. Gravitation is enormous at its surface, $2\frac{1}{2}$

times what it is here. A man weighing 10 stone here would weigh 25 stone there. But its density is very low, being one-quarter of that of the earth. It is a world of water and more or less dense gas.

Astronomers observe it with great interest since they see it furrowed with various currents and enveloped in clouds and vapours. The currents at the surface revolve with different speeds, so that it is difficult to know what is the rotation of the planet itself. At the equator, this rotation is accomplished in 9 hours 50 minutes 30 seconds; not far from the equator, in the subtropical regions, it takes 9 hours 55 minutes 41 seconds. Since Jupiter's year is twelve times as long as ours, there are more than 10,000 days per annum.

Here is a curious circumstance: on account of the difference of rotation, the Jovian year comprises 87 more (Jovian) days at the equator than it does in the subtropical zone. If it were the same on our globe, the inhabitants of the Congo, Colombia, Borneo, and Sumatra would have a day more in their year (and even five in four years) than those of Senegal, the Antilles, Siam, or India, a difference of 125 days per century! It would be difficult to keep their calendars in agreement.

There are seven or eight kinds of currents between the equator and the poles, so that one can say of Jupiter, as of the Sun, that it does not turn in one piece.

And those speeds themselves vary with the years. In the southern tropical zone there is a curious spot which has been followed with some interest for fifty years. Although it is much larger than the Earth, it seems to float in the current. Constant observation shows its period of rotation to have been 9 hours 55 minutes 41 seconds in 1900 and 9 hours 55 minutes 39 seconds in 1906, and its period returned in 1913 to the figure of 1900. What is the nature of this floating spot? In 1890 its colour was red. It then gradually got paler, and then pink once more. Its shape was that of a long oval, measuring 26,000 miles in length, or more than three times the diameter of our globe. The current in which it floats has not the same period of rotation as the spot itself. The spot is pushed from west to east, and it has shifted 57 degrees in two years. Now, a degree on the globe of Jupiter in that latitude represents 720 miles. This Jovian formation has, therefore, been displaced 41,000 miles, or a distance more than five times the

diameter of our planet. It is as if Australia were to detach itself from the bottom of the sea and float about on the surface of the Pacific Ocean! Does this oscillation, which we only see in plan, and not in elevation, indicate the formation of a satellite trying to disengage itself from its parent planet and not succeeding? It looks as if there were on this giant world of Jupiter no surface at all, but irregular aerial layers one over the other, and full of clouds. The temperature must be very high, and enormous masses of vapour are formed, to wrestle in prodigious storms. Though mythology is of no importance here, it is evident that Jupiter is indeed the god of thunderstorms. Jupiter is a world in the making, a sun which has lost its light, but not its heat. Its density, nearly equal to that of the Sun, is barely greater than that of water. A globe of vapour varied by mountains of clouds, impalpable Himalayas, aerial Alps in convulsions, fluid and constantly agitated Pyrenees! Its colossal bulk has prevented its cooling as fast as our Earth, and it will no doubt take tens of hundreds of millions of years to arrive at a temperature fit for inhabitants. In all probability there is nobody yet on Jupiter, neither men nor animals nor plants. This immense world is in its primordial state and prepares itself for the future.

Spectrum analysis shows that the substances which abound there are different from terrestrial substances. Any living beings who may develop there will probably be chemically different from terrestrial beings, and consist not of oxygen, hydrogen, carbon, or nitrogen, but of combinations of other substances. It will be a differently constituted world.

Our interest in observing it is not lessened by that consideration. The thinking mind knows that our present era is of no particular importance. It has been preceded by other eras and will be succeeded by other cycles. It is not because we live at this particular moment that our time can be said to be of any special value. The years contemporary with Jesus Christ or with the Pharaohs who built the pyramids had just as much value as ours, and when we shall no longer be there, the Earth will continue to revolve and measure out their days to our successors indefinitely. Whether Jupiter is inhabited this year or in 500,000 years or in 10,000,000 years is the same to the philosopher. Everything to him is present, both the future and the past ... this great subject has already been dealt with, and there is no need to refer to it again.

Let us greet Jupiter as the symbol of the future. Behind him, among the constellation, there are stars whose light, starting at the time of the siege of Troy, has only arrived to-day. Thus in the same celestial record the past and the future are thrown together, and they tell us that if there is anything interesting in human life it is Thought, the mind which contemplates the universe, which lives intelligently, and without which all Nature would be but the play of an automaton.

VII. HEARTBEATS ACROSS SPACE

CHAPTER VII

HEARTBEATS ACROSS SPACE

ACROSS 92 million miles of space the magnetic pulses of the solar heart are transmitted to the earth, and they make the small and light compass needle, ever trembling and seeking its pole, vibrate on its pivot.

This magnetic needle does not remain fixed in the magnetic meridian, but oscillates every day to the right and left of the line, i.e. to the east and west. The greatest deviation is produced at about 8 a.m. The needle stops and returns to the magnetic meridian of the time (it varies from year to year), crosses it a little after 10 a.m., and continues to deviate towards the west, reaching its greatest deviation about 1.15 p.m. It then returns to the meridian, which it reaches about 6 p.m. and crosses to the east. It moves very slowly, with a slight oscillation between 8 and 10 p.m. to the east, attaining at 8 a.m. the greatest easterly deviation with which we began.

Such is the daily oscillation of the needle, a process expressive of the unknown and mysterious vital current which traverses our planet and manifests, so to speak, the soul of the earth.

This phenomenon is absolutely general and is observed on the entire globe, from the equator to the poles, in the same manner; the amplitude of the oscillation increases with the latitude but not proportionately; it only amounts to 1 or 2 minutes of arc in the tropics, to 9 minutes in France, and 7 minutes in Norway. This variation corresponds sensibly to the variation of temperature, the amplitude of which increases from the tropics to the poles. Heat, electricity, water-vapour, and barometric pressure are all associated with it.

There are certain perturbations, to be mentioned presently, to which some human beings are sometimes a little too sensitive. We live enveloped in an invisible world.

The amplitude of this daily oscillation varies from day to day, from month to month, from year to year. If we take the mean of the observations of a whole year, we find that from one year to another it changes from a

certain amount to double that amount and that this annual variation is regulated by a law. It is periodic, and the average period is eleven or twelve years.

For instance, in 1870 the oscillation was twelve minutes of arc and in 1878 it was six minutes. I take the figures for London, Prague, Munich, Rome, and Milan—not those for France, for no observations were made in that terrible year, or they were made badly, and I made many enemies by insisting upon them.

That year, 1870, was a maximum. There were other maxima, but not quite so great, in 1883, 1893, 1905, and in 1916. The cycle varies from ten to thirteen years.

Now, this behaviour of the magnetic needle corresponds exactly to the state of health of the Sun, i.e. its activity or the number of sunspots or whirlpools which indicate it as well as its gigantic flames, protuberances, faculæ, and the various manifestations of its radiating power.

There was a very considerable maximum of sunspots in 1870. Another maximum occurred in 1883 and another in 1905, but the latter were less marked than that of 1870. In 1918 another very well defined maximum occurred.

Minima of sunspots, protuberances, and faculæ were observed in 1878, 1889, 1901, and 1913. Now, these maxima and minima correspond exactly with the variation of the compass needle, as is shown in the tables and curves published in my *Astronomie Populaire*.

If curves are traced for solar activity on the one hand and magnetic oscillation on the other hand, it is found that the two curves are absolutely parallel. Science has been able to trace this parallelism very far, as far back indeed as the eighteenth century, and it is found to include the aurora borealis, which is a magnetic phenomenon. The concordance is evident and absolutely incontestable.

Thus there constantly emanates from the Sun a force different from light and heat, a force we do not perceive with our senses and which places our small and mobile planet in constant communication with our central star, which is more than a million times greater in volume.

The Sun sometimes experiences violent expansions, perturbations, tempests, and magnetic storms. The smallest disturbances which happen in our formidable star are transmitted to us.

For instance: the years 1904 to 1908 were very strange so far as the Sun was concerned. Instead of the single maximum which is usually observed, there were two maxima, one in 1905 and another in 1907, 1906 being less active. The curve traced to represent the number of sunspots shows two peaks, in 1905 and 1907 respectively, with a depression in 1906. It is the same with terrestrial magnetism: maxima in 1905 and 1907, with a slight weakening in 1906.

When I discovered this unexpected coincidence for 1906, several sceptics were greatly surprised. Yet it is true.

It was the same when I accused the Sun of being the author of the interruption of telephonic and telegraphic communication on September 25, 1909, which took place over the whole of France, in all Europe, and throughout the world. The telephone girls were unjustly blamed and the telegraph engineers were equally innocent. Why had no newspaper thought of it? Perhaps because politics are cultivated more than science in the papers of our unbalanced days. I do not accuse our country more than another, the English, German, American, Australian, and other editors having been equally astray concerning the cause of the perturbation.

On that day the intensity of the earth-currents which produced the breakdown was 50 million amperes, whereas the instruments work regularly with 10 to 12 milliamperes. The whole terrestrial globe had been plunged into a magnetic field of great intensity, into a veritable dynamic ocean originating in a solar torrent. A large group of sunspots surrounded by faculæ had arrived on the eastern limb of the solar globe on September 17, and had gradually advanced towards the central meridian of the solar hemisphere turned towards us by the Sun's rotation. On the 19th, this group had become much larger. The spectroscope had indicated violent eruptions. The enormous group of sunspots was visible to the naked eye and numerous photographs of it had been taken at the Juvisy Observatory. It passed the central meridian of the Sun on the morning of September 24 and continued its course towards the western limb, disappearing from sight on October 1. The magnetic perturbation which struck our planet arrived 30 hours after

the passage of this group of sunspots across the central meridian of the solar hemisphere facing us.

We had already, on October 31, 1903, observed a similar cosmic phenomenon, and previously on November 18, 1882, and August 3, 1872. Even before that date a similar event took place on September 1, 1859. But memory is short, and in any case astronomical ignorance is universal.

The magnetic link, invisible but powerful, joins our Earth to the central body of the solar system. Pœbus Apollo holds us in the hollow of his hand at a distance of 92 million miles, and we feel his pulse as he feels our feeble heart-beats. It is not only gravitation, nor only light, nor yet only heat, which throws a celestial bridge from the Sun to the Earth; it is also electricity; it is also magnetism; it is a force still unknown and unexplained which no doubt maintains communication between all the worlds. For the ethereal wave touches Mercury, Venus, Mars, Jupiter, Saturn, Uranus, and Neptune as it touches us, and if we could utilise it we could communicate with our neighbours in the heavens.

Interstellar magnetism! At each stage in the advance of science new horizons open out, unexpected perspectives reveal themselves, enlarging the field of our conquest over nature, but what we do know already is a mere nothing in comparison with what we are yet to know.

What is the nature of the wave or substance which leaves the Sun and reaches the Earth to produce those agitations of the magnetic needle, those magnetic storms and auroræ boreales?

Contemporary physics has been busy for twenty years about ions, electrons, and electric particles. The great perturbations described above are supposed to be due to the arrival in the upper atmosphere of torrents of electrons projected by the Sun and impinging upon the Earth with a speed of several thousand miles per second. The phenomenon is most intense when these emanations from the day-star hit the Earth directly. When they pass on one side, nothing out of the way happens. It may be that the torrent projected in a certain direction by the Sun on September 24 filled space in the direction of projection, and that our globe, travelling along its orbit at 67,000 miles an hour, only passed through that region on the following day.

Ions and electrons, those convenient goblins of present-day physics, do they exist? Nobody has ever seen them. Perhaps they are only ingenious interpretations. What does certainly exist is electric force. We may also think of the repelling force exerted by the Sun, which blows so strongly upon the nuclei of comets and produces tails several million miles long and always in the direction opposed to the Sun. Electricity plays a great part in the appearance, the illumination, and the dislocation of comets' tails, as we see nearly every year in the photographs taken at the Juvisy Observatory and which make such picturesque and suggestive revelations (it even plays a part in organic cells and among microbes).

Whatever may be the real nature of the force in action, it is certain that the magnetic link differs from universal gravitation, but it is no less intimate and all-pervading. The twin forces of gravitational and magnetic attraction between worlds can only be likened to the universal force of love, which attracts souls towards each other.

VIII. IDEAS CONCERNING COMMUNICATION BETWEEN THE WORLDS

CHAPTER VIII

IDEAS CONCERNING COMMUNICATION BETWEEN THE WORLDS

THAT in a future, which is perhaps not very far distant, say in a century or two, the inhabitants of our planet might enter into optical, electrical, or telepathic communication with those of another planet of the solar system is an event which we have some right to expect, though it is not a matter of to-morrow.

About the year 1840 the astronomer von Littrow, Director of the Vienna Observatory, put forward the idea of attempting an optical communication with the Moon. A triangle traced upon the soil of the Moon by three luminous lines each from 7 to 9 miles long would be visible from here by means of our telescopes. We even observe much smaller detail, such as the singular topographical designs noticed in the lunar circle which goes by the name of Plato. Thus a triangle, a square, or circle of that size, constructed by us on a vast plain by means of luminous points, either during the day by the reflection of sunlight, or during the night by means of electric light, would be visible to the astronomers of the Moon, if such astronomers exist and if they have optical instruments equivalent to ours.

The rest of the argument is of the simplest. If we observed on the Moon a correctly constructed triangle, we should be considerably interested. We should ask ourselves whether we had observed wrongly or whether an accidental geographical movement might have given rise to a regular figure. We should no doubt end by admitting that exceptional possibility. But if we suddenly saw that triangle change into a square, and some months later be replaced by a circle, we should then admit logically that an intelligent effect proves an intelligent cause, and we should think with some reason that such figures undoubtedly reveal the presence of geometers on the neighbouring world.

From that position to a search for the reason for the formations on the Moon's surface and to the question why and with what object those unknown comrades drew those figures would be a short step quickly taken. Would it be with the idea of entering into communication with us? The hypothesis would not be absurd. It would be put forward, it would be discussed, it would be

rejected as far-fetched and defended as ingenious. And after all, why not? Why should the inhabitants of the Moon not be more curious than we, more intelligent, of higher aspirations, less stuck in the bog of material necessities? Why should they not suppose that the Earth can be inhabited as well as their own world, and why should these geometrical appeals not be made with the object of asking us whether we exist? Besides, it is not difficult to reply. We are shown a triangle, we reproduce it here. A circle is traced, we imitate it. And so communication is established between the Heavens and the Earth for the first time since the beginning of the world.

Geometry being the same for all the inhabitants of the universe, and twice two making four in all the regions of infinite space, and the three angles of a triangle always being equivalent to two right angles, signals thus exchanged between the Earth and the Moon would not even have the obscurity of the hieroglyphics deciphered by Champollion, and the communication, once established, would become regular and fertile. Besides, the Moon is close by. Its distance of 240 thousand miles is only thirty times the diameter of the earth, and many a rural postman has covered that amount of ground on foot during the course of his life. A telegram would arrive there in a second and a quarter, and light takes the same time to cover that distance. The Moon is a celestial province joined to our destiny by Nature herself.

Up to now we have noticed nothing on the Moon which might lead us to suspect the existence of a thinking humanity inhabiting that small celestial island. Tet the astronomers who specially observe our satellite and who study its singular aspects with attention and perseverance are generally of opinion that that heavenly body is not as dead as it seems. We must not forget that in the present state of optical science it is difficult in practice to apply to the study of the Moon a magnification greater than a thousand times. To see our satellite a thousand times closer than it really is still leaves it at a distance of 240 miles. Now, what can be distinguished at such a distance?

It is certain that mysterious variations actually occur on the surface, notably in the arena of the amphitheatre of Plato as mentioned above. What is also certain is that the lunar globe, 49 times smaller than the earth and 91 times lighter, exerts on its surface a gravitational attraction six times less than that which exists at the surface of our own planet, so that an atmosphere analogous to the air we breathe would be more rarefied and difficult to perceive from here. It is therefore not surprising that this neighbouring world is so different from ours. Besides, seen from a balloon 3 or 4 miles above ground, the Earth

appears deserted, uninhabited, as silent as an immense cemetery, and whoever arrived from the Moon in a balloon could ask himself even at that small distance whether there were any people in France or any sound in Paris. Such at least is the impression conveyed to me by my aerial voyages.

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The cold and deathly aspect of our pale satellite was not encouraging for the realisation of the original project of the astronomer J. von Littrow, and soon afterwards, forgetting our nearest neighbour, the imagination of some physicists was bold enough to consider the planet Mars, which is never nearer than 35 million miles, but is the best known of all the countries of the sky and which offers so many points of resemblance with our world that we should hardly feel exiled if we were to transport our household gods thither. The aspect of Mars indeed tends to comfort us after that of the Moon. One could imagine oneself in some terrestrial region. The seasons, summer and winter, autumn and spring, days and night, mornings and evenings, waters, clouds, snows, atmospheric variations, plains adorned with various vegetations—all these things present many resemblances to our own world. The years are longer there since they last 687 days, but the extreme variations of the seasons are about the same as with us since the inclination of the axis is about the same. The days are also a little longer, since the diurnal rotation is 24 hours 37 minutes 23 seconds, but the difference is not great. And you will notice that this is all known with great precision, that diurnal rotation, for example, is known to within one-tenth of a second, or we might even say one-hundredth of a second.

When, on a fine starry night, we examine this world through the telescope, when we see the polar snows which melt in the spring, the finely marked continents, the mediterraneans with long gulfs, the eloquent and varied geographical configuration, one cannot help asking whether the Sun which illuminates that world as it does ours shines on nothing living, whether those rains fertilise anything, whether that atmosphere is breathed by any living being, and whether that world of Mars which rolls swiftly through space resembles a railway train travelling empty without either goods or passengers. The idea that the Earth where we are could swing round the Sun as it does without being inhabited by any creature whatever appears so absurd as to be hardly worth thinking about. By what permanent miracle of sterilisation would

the forces of nature which act there as they do here have remained eternally inactive and barren?

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It is therefore conceivable that one might apply to Mars the plan originally proposed for the Moon. The distance of that planet is such that although it is much larger than the Moon in volume, it appears 63 times smaller when it is nearest to us. Yet one may understand that a telescope which magnifies 63 times enlarges Mars to the size of the Moon as seen with the naked eye, and that a magnification of 630 gives it a diameter ten times larger than that of our satellite.

Yet if ever the attempt is made to put any sort of project of communication between us and Mars into practice, the signals would have to be carried out on a much vaster scale. It would not be triangles, squares, circles, of several miles which would have to be constructed, but figures of 70 miles or more, always supposing (1) that Mars is inhabited, (2) that these inhabitants occupy themselves with astronomy, (3) that they have optical instruments analogous to ours, and (4) that they carefully observe our planet, which to them is a brilliant star of the first magnitude, the morning and evening star, and in fact the brightest heavenly body.

Is this fourfold hypothesis acceptable! If that question were put to the vote of the citizens of the Earth, without asking the opinion of the Central African savages or the South Sea Islanders, but only the numerical majority of the European population, one may safely wager that they would not even understand the question, for the majority of mankind does not know that the Earth is a planet and that the other planets are Earths.

And then there is sound common sense which reasons so justly on account of the excellence of its education. "We are," it says, "without doubt the most intelligent beings of creation. Why should other planets have the honour of being enriched by intellectual excellence such as ours? Can we even admit the existence of beings similar to ourselves?" No doubt one could perhaps remark that the most gifted nations of the Earth do not know how to conduct themselves, that their intelligence is chiefly exercised in devouring each other and in ruining themselves, that they mortgage the future like blind fools, that thieves are not uncommon, nor even murderers. But apart from that we are obviously very superior beings, and it is hardly likely that on any of the

myriads of worlds which gravitate in the immensity of space nature should have given birth to intelligences of our calibre.

Why therefore should we attempt an optical correspondence with the planet Mars? If it is inhabited, the inhabitants have not our powers, and our trouble would be thrown away. Even should they see our signals, they would never think that they were meant for them.

Also, we shall never attempt it.

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But is it possible that the inhabitants of Mars have already commenced and that it is we who do not understand?

According to geological computations, the minimum age of the Earth as a habitable sphere since the formation of the first dry land is 20 million years: 10 million 700 thousand for the primordial age, 2 million 300 thousand years for the secondary age, 460 thousand years for the tertiary age, and 100 thousand for the quaternary age. Man has existed on Earth since the end of the tertiary age, that is over 100 thousand years. Astronomical instruments were only invented in 1609, and Mars was not observed nor recognised in its principal geographical details until 1858. Complete observations of Martian geography only date back to 1862. The first detailed triangulation of the planet, the first map comprising the smallest objects visible in the telescope and measured by micrometer was only commenced in 1877, continued in 1879, and completed in 1882. It is therefore only a few years since Mars entered into the sphere of our practical observations. I may add that only very few of the Earth's inhabitants have seen it in all its details, one of the foremost of these being the astronomer Schiaparelli, the Director of the Milan Observatory.

According to the most probable cosmogonic theory, Mars is older than our planet by several million years and much more advanced along its path of destiny than ours. The inhabitants of Mars may have been signalling to us for 100 thousand years. Nobody on our planet would have suspected it. It is only since the seventeenth century that astronomers may have thought, not of discovering such signals, for their instruments were not sufficiently powerful, but of the possibility of some day seeing a little more of what happens on that neighbouring world. In fact it is only a few years that we have had any hope of distinguishing these minute details, not to speak of explaining them.

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Now, this is what happened. The map of the planet Mars was made with infinite care by the able astronomer of Milan. One sees on this chart in several regions certain points where the observer has found the presence of luminous patches shining like snow illuminated by the sun. That these luminous patches are due to snow is not likely, because some are seen near the equator in the tropics as well as at higher latitudes. Nor does it seem likely that they are the tops of mountains, for they are close to the seas and arranged symmetrically to certain straight canals. Besides, several of them seem to trace out meridians and parallels of latitude, and on seeing them one involuntarily thinks of Geodesic signals. One sees triangles, squares, and rectangles.

That these luminous points should have been established by engineers or astronomers on Mars is not my idea. That the 60 great rectilinear parallel canals which we admire on that planet and which establish communication between the Martian seas should be the work of the inhabitants of this neighbouring planet would be presumptuous to imagine. That is not the conclusion I would arrive at. Nature is so rich in procedure, so varied in its manifestations, so multiple and so complex in its effects, and often so strange and original in its play, that we have no right to limit its mode of action.

But it is none the less true that if the inhabitants of Mars wished to send us any signals, that method of procedure would be one of the simplest and it is in fact the only one which has been devised among ourselves. They could not do better than thus dispose luminous points at certain distances according to geometrical figures. For instance, one finds the intersection of the 267th meridian with the 14th degree of northern latitude a region limited by points situated at distances from each other corresponding to Amiens, Orleans, and Le Mans. If the inhabitants of Mars wished to send us signals they could not have chosen better places for their luminous stations.

I am far from saying that this is so and that there is any intention in this arrangement. Yet if it were so, it would be we who did not understand.

And there is nothing surprising in this. The inhabitants of the Earth take no interest in the Heavens. Ninety-nine per cent, of them among the 1,500 million Earth-dwellers do not know what they walk on and have no conception of the reality. They are busy with eating, drinking, and reproducing themselves, with amassing various objects, with patriotically devouring each other, and with

dying. But as regards asking where they are or what is the universe, that is no concern of theirs. Their native ignorance suffices them. They live in the middle of the Heavens without knowing it and without the slightest enjoyment of intellectual happiness which some select spirits find in the recognition of truth.

The inhabitants of Mars, on the other hand, being more ancient than ourselves, can be much more advanced in the way of progress and can live an enlightened and intellectual spiritual life. We are safe in supposing that they know our world much better than we know theirs, and that our astronomical science is only child's place beside theirs. If, therefore, the people of Mars, living perhaps for a long time already in the harmony of a peaceful and intelligent life, had thought of attempting to send signals to the Earth, with the idea that perhaps our planet is also inhabited by an intellectual race, then since they have never received any answer from us they will have concluded that we are not on their level, that we do not busy ourselves with the matters of the sky, that astronomy and optics are not very advanced among us, and that in all probability we have not yet emerged from clumsy material instincts. Is their conclusion very far from the truth?

Perhaps also the Martian academies declare the Earth uninhabitable and uninhabited: (1) because it is not absolutely identical with their planet, (2) because we have only one Moon whereas they have two, (3) because our years are too short, (4) because our sky is very often overcast, whereas theirs is almost always clear, (5 and 6) for a thousand other reasons, each as cogent as the rest.

As we have seen above in the chapter on Mars, one often notices besides the luminous points we have mentioned certain more extensive brilliant projections which appear on the terminator and which must be caused by the reflection of the rising or setting sun on snowy peaks or on clouds which are certainly not signals. However this may be, of all the bodies which blaze in the skies on a clear night, and particularly of all those bodies which gravitate with us round the Sun, there is one which engages with a captivating interest the attention of astronomers. It is the singular little world of Mars. But it is not so easy to communicate with it as was supposed by that good lady of Pau, Madame Guzman, who left a legacy to the Académie des Sciences in the shape of a prize of 100,000 francs to be given to the first person who should discover a means of communicating between the Earth and another world—with the exception of the planet Mars, because it would be too easy! So true it is that on our planet the best intentions are often mixed up with a few grains of folly.

After steam, the electric light and the telegraph and the telephone, the discovery of unmistakable signs of the existence of humanity inhabiting another region of our solar archipelago, would that not be the most marvellous apotheosis of the scientific glory of the twentieth century? And cannot wireless telegraphy be some day applied to this problem? Electro-magnetism is an immense invisible world still insufficiently explored. Let us wait, observe, and study.

IX. STARS AND ATOMS

CHAPTER IX

STARS AND ATOMS

LAST night in the calm silence of midnight, while all nature slept, I observed in the telescope a small fixed star lost in the multitude of the heavenly host, a pale star of the seventh magnitude separated from us by an almost immeasurable distance.

In my thoughts I travelled up to it. I remembered that this star is not visible to the naked eye; that there are 19 stars of the first magnitude, 60 of the second, 182 of the third, 530 of the fourth, 1,600 of the fifth, and 4,800 of the sixth magnitude (which gives a total of about 7,000 stars visible to the naked eye in the case of persons gifted with acute vision); but that the stars of the seventh magnitude, one of which I was observing, are 13,000 in number, and those of the eighth magnitude 40,000; that the number grows progressively as we ascend beyond natural vision. I remember that the sum of the stars of the first ten magnitudes amounts to 560,000, that of the first twelve magnitudes to 4 million, and that we reach more than 40 millions in taking the first fifteen magnitudes.

Without losing myself in the profundity of infinite perspectives, I concentrated my thought as I had already concentrated my gaze upon this simple seventh-magnitude star of the constellation of the Great Bear, which never descends below the horizon of Paris, so that we can observe it almost any night in the year, and I remembered that it shines 200 billion miles from here—a distance which an express train rushing along at 75 miles an hour would take no less than 325 million years to traverse.

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Transported to such a distance, the glorious sun which illuminates us would lose its splendour and its glory. Not only would it be invisible to the naked eye, not only would it fail to contribute to the brightness of the midnight sky, but it would fall considerably below the star of the seventh magnitude above mentioned and would only be discoverable by the most careful telescopic search. That small star, therefore, which is only a small brilliant point in the starry heavens, is in reality an immense sun of giant size, greatly in

excess of that on which our planet depends for its light. The latter is already 333,000 times heavier than the Earth and 1,300,000 times more bulky. In claiming for that small star a weight more than a million times that of our globe and a volume equalling that of several million Earths we should be well within the region of possibility.

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These views in connection with a simple star almost forgotten among the multitude of its sisters bring us face to face with the most formidable realities of the constitution of the universe. But they do not yet represent the most interesting aspect of our contemplation. It is a singular fact, quite unexpected by the ancient philosophers, fantastic and hardly conceivable to the seeker after truth who endeavours to comprehend its real value: It is that these suns of the infinite, far from being fixed as they seem to be on account of their immense distance, are travelling through space with tremendous speed. The star in question^[2] runs, flies, hurls itself across the immensity with a speed of about 20 million miles per day!

There are no fixed stars.

Yes, 6,800 million miles per annum! And yet, in ten years, in fifty years, in one hundred years, this star will be barely displaced in the sky! The speed of a bullet or a shell fired by one of our most powerful guns being only 2,340 feet per second and that of this star being a million feet per second, we see that the speed of the star surpasses that of the shell in the proportion of 457 to 1! Can the most daring imagination picture such a flight!

In five days and some hours the star would cover the 92 million miles which separate us from the Sun, a distance which a gunshot would take seven years to cover. It is clear that such a speed is marvellous, yet it exists, and has been measured by delicate and precise operations. It cannot in any case be less than the figure we have indicated.

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This speed is a symbol, and it is as such that I wish to present it here. All the stars are endowed with such movements more or less rapid, and not only all the stars—each of them a sun and most of them the centres of planetary systems, dispensers of light, heat, and harmony, around which gravitate habitable earths, the present, past, or future abodes of various beings and

terrestrial things—not only, I say, are all the stars thus driven through the void, but all the planets, all the satellites, all the worlds and systems, all creation.

The Earth swings round the Sun carried along with a speed of 1,600,000 miles per day, meanwhile rotating on its axis, animated by a dozen different sorts of movements, lighter and more mobile than a child's balloon floating in the air, solicited by the attractions of various neighbouring bodies, a veritable plaything of cosmic forces which carry us along in an immense whirlpool. The Moon revolves round the Earth, constantly deranging us in our progress and subjecting us to perpetual undulations. The Sun carries us along with all its cortège towards the constellation Hercules, so that since its very origin our Earth has never passed through the same place twice, but describes in space, not closed ellipses, but spirals which roll on without end. The suns adjoining ours move with their systems in various directions. The constellations are dislocated from one century to another, each star being animated by a proper movement which produces a constant modification of the figures in the sky. In addition to all this, our whole sidereal system is carried along in space by a joint movement. And thus everything moves, everything circulates, everything rushes, with breathless speed, towards a goal which is unknown and which is never reached.

This is not a romance, a dream of pure contemplation, an outside view of ourselves: it is our own history, fatal and inevitable. Since an hour ago, every one of us, reader or writer, rich or poor, wise or ignorant, infant or greybeard, whether we are active or asleep—since an hour ago every one of us has described in the sky an invisible track of more than 60,000 miles, for our planet traces out 580 million miles per annum merely by its revolution round the Sun, and a centenarian lives to mark out a distance of 58,000 million miles. Now, it is found that these speeds are the very condition of the stability of the universe: the heavenly bodies, the Earth, planets, satellites, suns, stellar systems, star clusters, galaxies, and remote universes, are mutually sustained by the equilibrium of their attractions. They are all suspended in the void and maintain themselves in their ideal orbits simply because they turn quickly enough to create a centrifugal force equal and opposite to the attraction exerted upon them, so that they remain in an unstable but perpetual equilibrium.

Our Sun carries us along towards the constellation of Hercules: this has been known for a hundred years; but we know now that the constellation of Hercules forms part of our sidereal universe and that that universe is travelling

in a certain direction. We therefore only perceive relative movements. Whither are we going? Vain question. We are going—into infinite space.

Formerly people were troubled, not without reason, concerning the solidity of the foundations of the world, for before the isolation of our planet in space and its movement round the Sun had been demonstrated it seemed indispensable to give to the Earth an unshakable foundation and an unlimited route. But since the heavenly bodies rise and set and pass under our feet this foundation had to be given up, apart from the fact that it did not satisfy the most far-seeing minds. It is quite impossible for us to conceive a material pile, however solid or thick, even as thick as the Earth's diameter and rooted in infinite space, just as one cannot imagine the real existence of a stick which has only one end. However far the mind can descend towards the base of that material pillar, the end must come at last, since only empty space can be without end, and then that terrestrial pillar serves no purpose at all since it is itself without a support. Besides, travellers succeeded one day in voyaging all round the globe, and nowhere was this imaginary pillar discovered.

The modern conception of force in contrast with the ancient idea of matter has a philosophical import without precedent in the whole history of the science. It teaches, proves to us, and convinces us that the visible, palpable, material universe rests on the invisible and the immaterial, on imponderable force.

That is a fact against which the misleading testimony of the senses can no longer prevail. The Earth, which was believed to be the stable basis of creation, is itself not sustained by anything material, but by invisible force. The void extends above and below, right and left, and to infinity in every direction. It is sustained by solar attraction and by its own movement. The same applies to all worlds, to all heavenly bodies, to everything which composes the universe, to the intimate constitution of bodies as well as the sidereal total. The Earth, the planets, the suns, the stars, the stellar systems, are the mobile atoms of the grand organism of the universe. The Milky Way is a dust in which every grain is a sun.

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From the infinitely great, let us now descend to the infinitely small

The substances which appear to us most solid and hard are composed of molecules which do not touch each other. Every one of these molecules is

invisible to the naked eye and is itself made up of still smaller atoms which do not touch either.

A bar of iron, for example, is composed of molecules which do not touch, which are in perpetual vibration, which separate under the influence of heat and close up under the influence of cold. Exposed to the Sun, the temperature of that bar reaches about 60 degrees centigrade; cooled by the ice of winter, it descends a few degrees below zero. Now, the length of that bar varies between the first condition and the second, and its molecules can be further separated by heating them to a higher temperature: they can thus be so far separated from each other that they exercise no further mutual attraction. When that happens, the bar melts and forms first a liquid and then a gas.

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The Eiffel Tower is a little higher in summer than in winter and in the afternoon than the morning, on account of the variation produced by solar heat. The difference in height can attain 6 inches.

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The smallness of molecules surpasses anything one can imagine. In the gold-beating industry gold-leaf has been obtained so thin that a quarter of a million leaves go to the inch. Each gold-leaf, therefore, has only a thickness of a 1/10,000 of a millimetre. Now, it is composed of molecules a considerable number of which is required to fill up that thickness. Two hundred of them arranged in a line and separated by intervals equal to their own diameters would just fill up the thickness of a gold-leaf.

In experiments made on the action of oil upon the surface of water for calming the waves, it has been found possible to cover 400 square yards on the Lake of Geneva with 20 c.cm. of oil, which reduces the layer of oil also to 1/200,000 of a millimetre. Supposing that under these conditions the molecules of oil are in contact and in a single layer, they would at the most have that diameter.

By mechanical means a millimetre has been divided on a plate of glass into a thousand equal parts. There are animalculæ so small that their whole body placed between two of these divisions does not touch them. These living beings then measure at the most 1/1,000 of a millimetre, or what is nowadays called a micron. They have members, organs, muscles, nerves, etc. These

organs are composed of cells and the cells of molecules. If the latter were only a hundredth part of the dimensions of the body (they are probably much smaller), these molecules would measure, if separated by intervals equal to their own size, 1/200,000 of a millimetre as before.

Molecules or atoms, whatever may be the name applied to the ultimate particles of matter, are declared by present-day science to be equal to stellar systems or microcosms.

* * * * *

This modern teaching of microscopy was anticipated for a long time by thinkers.

In his *Commercium Philosophicum*, published in 1745, Bernoulli wrote to Leibnitz concerning the imaginary inhabitants of a grain of pepper.

“If these animalculæ had an intelligent mind and were capable of reasoning, they could flatter themselves that they and the drop of liquid which they inhabit constituted the entire universe. Imagine that a small grain of pepper in which under the power of the microscope we discover a million animalculæ had all its parts proportional to the corresponding parts of our world, that is to say, its sun, its fixed stars, its planets with their satellites, its earth, and its mountains, its fields, its forests, its rocks, its rivers, its lakes, its seas, and its diverse animals, can one suppose that the inhabitants of the grain of pepper, these *pipericals*, who would perceive all objects under the same visual angle and consequently in the same size as we do, would not believe that outside their grain nothing exists, and would have the same right to believe it as we believe that our world includes everything? For, I ask you, what reason or what experience would they have which would convince them of the contrary or would show to these poor little animals that there is another world incomparably greater than their own with inhabitants incomparably greater than themselves?”

Therefore, concludes Bernoulli, if these *pipericals* cannot know that, who is there among us who knows that this whole visible world is not perhaps a grain in comparison with another world incomparably greater?

The learned geometrician of Basle summarises his idea as follows: “I believe that there can exist in nature other animals who are in size as far above us and ordinary animals as we and our animals are above microscopic animalculæ, and who observe us in our world with their microscopes as we

observe that infinite multitude of animalculæ with ours. I go further and maintain that there might be beings incomparably larger again, and I suppose as many degrees upward as are found in going downwards, for I do not see why we should constitute the highest degree.”

Leibnitz replied to Bernoulli: “I am not afraid of advancing the opinion that there are in the universe animals that are as much our superiors in size as we are above the animalculæ which we only discover by means of a microscope, for nature knows no limits. On the other hand, it may be and it must be that there are in the small grains of dust and in the smallest atoms worlds which are not inferior to ours in beauty and variety.”

The exceedingly small microscopic organisms discovered and studied by Ehrenberg at the beginning of the nineteenth century seemed to place this population in evidence.

Pascal had written, as early as 1660, or perhaps even in 1654, in his *Pensées*:

“What is man in the infinite? Let a flesh-worm offer him in the smallness of its body parts which are yet smaller. Legs with joints, veins in those legs, blood in those veins, humours in that blood, drops in those humours, vapours in those drops. Suppose that in dividing up these last things he should spend his forces of imagination and that the last object at which he could arrive be the subject of our present discussion. He will perhaps think that that is the extreme of smallness in Nature. But I shall open for him a new abyss. I shall not only paint for him the visible universe but the immensity of nature which one can conceive within the range of this small atom. Let him see in it an infinity of worlds, every one with its firmament, its planets, its earths, in the same proportion as the visible world, on that earth animals and finally flesh-worms in which he will find again what the first flesh-worms presented to him. Let him go on finding the same thing without end and without rest. Let him lose himself in those marvels as astonishing in their smallness as the others are in their size, for who would not be astonished that our body, hardly perceptible in the universe which itself would be imperceptible in the grand totality, should now be a colossus, a world or rather a universe in comparison with the nothingness which we can never arrive at.

“Whoever follows these thoughts will be afraid of himself and, considering himself sustained by the weight which nature has given him, between those two abysses of the infinite and nothing, he will tremble in face of those

marvels, and I believe that as his curiosity changes to admiration, he will be more disposed to contemplate them in silence than to seek them with presumption.

“For indeed what is man in nature? A nothing in comparison with the infinite, a universe in comparison with nothing, a mean between nothing and all. Infinitely removed as he is from comprehending the extremes, the goal and principle of things are invincibly concealed from him in an impenetrable secret; he is equally incapable of seeing that nothingness whence he sprang or that infinite which swallows him up.”

More than one of our contemporary savants has republished this conception of “the atom as a world system,” imagining that it is new and without appearing to recollect either Bernoulli or Pascal.

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As we have seen before, the invisible world is the basis of creation and the visible universe is composed of invisible bodies. What we see is made up of things which we do not see.

In the sky every star of the Milky Way being below the seventh magnitude is quite invisible to our eye. Yet we see the Milky Way.

On the Earth we see and touch crowds of molecules every particle of which could be neither seen with our eyes nor felt with our touch nor perceived with any of our senses. And yet one meets very often so-called scientific men in the world who say peremptorily: “I only believe what I see.”

A large number of different measurements of the speed of light agree in proving that this speed is 186,000 miles a second. That is a fact of observation absolutely proved by astronomy and by physics.

These 186,000 miles amount to 982 million feet or 11,785 million inches.

Now, certain measurements of the same degree of precision show that in a ray of red light there are 42,300 waves to the inch or 1,666 to the millimetre.

Each of these waves therefore measures $\frac{6}{10,000}$ of a millimetre. Those of the extreme red measure $\frac{7}{10,000}$; beyond the red, wave-lengths of $\frac{8}{10,000}$ or $\frac{9}{10,000}$ are found; the blue waves measure $\frac{4}{10,000}$ to $\frac{5}{10,000}$ and the violet $\frac{4}{10,000}$.

Expressed in millionths of a millimetre, as is usual in spectroscopy, these wave-lengths are the following:—

WAVE-LENGTHS OF LIGHT

Infra-red	900 or more
Extreme-red (A)	760
Red (B)	687
Orange (C)	666
Yellow (D)	689
Green (E)	617
Blue (F)	486
Indigo (Q)	431
Violet (H)	897
Ultra violet	380 and less

(These figures are those of the solar spectrum, measured for the first time by Fraunhofer a century ago.)

Those are waves in the ether, which is in perpetual movement.

If we multiply 300,000 million millimetres by 1,666, we find the number 500 billion. That is the number of waves which enter our eye in one second when we look at red light.

It requires 2,230 waves of violet light to fill up a millimetre and 700 billion shocks per second to give the sensation of that colour.

Between the red and the violet all the other colours range themselves. The vibrations which give the impression of red are the slowest and their waves are the longest.

This subtle substance, ether, penetrates all the bodies; it surrounds the most minute atoms of objects and beings, solids and liquids.

Studies in molecular physics have led to this conclusion, that in a cubic centimetre of air the molecules which compose it only occupy a third of a cubic millimetre, that is to say, the $\frac{3}{1,000}$ part of the total volume. It is like a

cathedral in which children's balloons might float, the remaining space being empty.

All these molecules, all these systems of atoms, are in perpetual motion like the worlds in space, and the structure of bodies is organised by invisible force. In hydrogen, at ordinary temperature and pressure, every molecule is endowed with a speed of translation, vibration, and circulation, of more than a mile per second.

Every body, organic or inorganic, air, water, plant, animal, man, is thus formed of molecules in movement.

The analysis of bodies, both organic and inorganic, therefore brings us into the presence of movements of atoms controlled by forces, and the infinitely small speaks to us the same language as the infinitely great.

The powerful microscopes of to-day and instruments of projection show in these microbial movements in these cells a fantastic life such as that of organisms which circulate by the thousand million in our own blood.

The molecule, intangible and invisible, and hardly imaginable by our minds accustomed to superficial judgments, constitutes the only true matter, and what we call matter is, singularly enough, the effect produced upon our senses by the movements of molecules, constituting as it were an incessant possibility of sensation.

It follows that matter, like the manifestations of energy, is only a mode of motion. At the temperature of absolute zero, matter, as we know it, would cease to exist.

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The name of materialist, still borne to-day by certain people who see no farther than the ordinary appearances of things, could only be considered by the thinker as an outworn expression without meaning. The visible universe is not at all what it appears to be to our senses: it is the invisible universe which constitutes the essence and basis of creation. In fact, this visible universe is composed of invisible atoms which are not in contact; it hangs in the void, and the forces which control it are themselves invisible and immaterial. You may look for matter, but you will not find it; it is a mirage which recedes as we advance towards it; it is a spectre which disappears whenever we seem to seize it. It is not the same with force, the dynamic element; it is invisible and

imponderable force that we find in the last analysis and which represents the basis, the support, and the very essence of the universe.

In the deep and silent night everything moves driven by the breath of God. In these hours of calm reflection, do we not hear the voices of the infinite? Night is the normal condition of empty space, and we only have day during a half-rotation of the Earth because we dwell in the immediate neighbourhood of a star. Night fills all; but it is not darkness; it is the soft light shed by millions of stars. We can thus better appreciate how everything is in vibration. The movements of every atom on earth and in the heavens are the mathematical result of all the ethereal undulations which arrive in time from the abysses of infinite space. The moon attracts the Earth, the Earth attracts her sister-planets, these beckon and call her, the stars attract the Sun; and, like those motes of dust which one sees oscillating and vibrating in a beam of light, so also glide, turn, circulate, fly, vibrate, and palpitate all the worlds and all the universes, out to infinity, amidst limitless and bottomless space.

A geometrician has dared to say that by stretching out his hand he could disturb the Moon in its course. This is a vivid expression of the extreme mobility of things intended to show that the feeblest displacement of the centre of gravity has a far-reaching effect. When the Moon passes over our heads it raises the whole Earth, displaces the waters of the ocean, and makes every one of us weigh 18 milligrammes less than when it is on the horizon. When Venus passes at 25 million miles from here or when Jupiter passes at 375 million miles, both displace our whole planet from its normal position.

Have you ever brought a bit of iron near a freely suspended magnetic needle? What a marvellous spectacle is offered by the mobility, the oscillations, the mad rushes of the needle under the influence of an object apparently inert which acts upon it at a distance. We observe a compass needle at the bottom of an hermetically sealed vessel: a regiment passes on a neighbouring road and the needle becomes agitated, influenced at a distance by the steel bayonets. An aurora borealis appears in Sweden, the compass feels it in Paris; nay, we have seen above that the fluctuations of the magnetic needle are in relation with the spots and eruptions on the Sun. The new physics is the proclamation of the invisible universe.

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It is under this aspect which it appeared to me interesting here to contemplate the visible universe, inviting to this contemplation those among my readers who wish sometimes to think of profounder truths. Stars and atoms bring us face to face with an immense symphony. Those who only see the orchestra without hearing anything are the deaf. Behind the visible world, our minds must feel the presence of the invisible world upon which we are based. All that we see is appearance: the real is the invisible, the force, the energy, which moves all and carries all through infinity and eternity.

And indeed we are really in the infinite and the eternal. The little star of which we spoke before, an enormous sun a million times the size of the Earth moves at such a distance that the fastest express train would take at least 325 million years to reach it. Yet it is one of our neighbouring stars. One can go still farther and farther and proceed with any speed through any number of centuries in any direction of space without ever coming to an end, without ever advancing a single pace, since the centre is everywhere, the circumference is nowhere, and eternity itself does not suffice to vanquish infinity.

FOOTNOTES:

[\[2\]](#) This star is called “No. 1830 Groombridge.”

X. ARE OTHER PLANETS INHABITED?

CHAPTER X

ARE OTHER PLANETS INHABITED?

WE have a tendency to remain geometric and anthropocentric and to believe that everything is created on the terrestrial model. Not long ago, at one of the scientific soirées of the Solar Festival which I founded in 1904, and which is almost always illustrated by a conference of learned men of philosophical attainments, I requested my eminent friend Edmond Perrier, Member of the Institut and Director of the Paris Museum, to discuss the question of the population of the planets in the light of the latest achievements of science, in which he is past master. His reasoning was, that the same matter, the same forces, the same laws, exist on the Earth, Venus, Mars, Jupiter, etc., and that therefore the evolution of life is everywhere the same, arrested in one place and developed in another according to the conditions and circumstances, and that all organisms on all the planets can only have terrestrial forms. Our palæontology would be repeated everywhere.

It seems to me that this idea, shared, by the way, by other learned naturalists, is really too “naturalist,” too terrestrial, too classical, too professional, too narrow, too little in harmony with the grandeur of the universe, with the immensity of its energy and the variety of vital manifestations found on our small planet. On the contrary, the diversity of beings, already so prodigious on our little globe, must be, so to speak, infinite, and the extra-terrestrial living forms cannot be cast in the same mould. There is no reason, for instance, why all the beings of the universe should be limited to our five senses. There are inevitable differences: gravitation, density, food-supply, atmosphere, temperature, light, the years, seasons and days, etc., etc.—causes so different cannot fail to produce absolutely different effects.

A savant is a man accustomed to discussions and delighted to provoke them, because he knows that they contribute to the advancement of science; the naturalists whose views I dispute will not bear me any ill-will, and this is not the first time that they will pardon me for being a recalcitrant microbe.

The philosophers who teach that the universe is both infinite and homogeneous resemble microbes who think that their cell is the universe. Let us imagine the microbes in a particle of rust attempting to reason on the life of

iron according to their own observations. For them, the world is a bit of iron attacked by oxygen. There is nothing else in nature. All their science leads them to conclude that the universe is made of iron. If they could have any vague notion about the existence of grass, insects, men, the sun, Jupiter or Sirius, they would be firmly convinced that they were all made of iron. If, in their particle of rust, they had observed the movements of translation and rotation in the constituent atoms of iron and could have risen to astronomical notions beyond their sphere, they would conclude that the other planetary systems are also made of iron and that none but iron-dwellers can exist. If we suppose that not only our own solar system but our visible universe and the other universes which succeed each other without end in the depths of infinite space are all constructed on the same zoological plan, we reason like these microbes.

Since the publication, a long time ago (1862), of my first work, entitled *The Plurality of Inhabited Worlds*, in which I expounded and discussed the conditions of habitability of the planets of our system as known, to science at the time of publication, several astronomers and philosophers have taken up the same question of the different aspects. Among these studies there is one which seems particularly worthy of attention, recently written by Professor Scheiner, Director of the Potsdam Observatory. I here offer my readers a condensed translation, asking them to excuse the Germanisms which render the style a little rough. But however bitter the rind, the fruit is good.

GENERAL REMARKS

“The aspect of the starry heavens during a clear and calm night gives a joy which only superior souls can feel. In the bosom of the general silence of nature and in the calm of all our senses our immortal soul whispers an undefinable language and forms conceptions which are difficult to express. If among the thinking beings of our planet there are vulgar spirits which remain slaves to vanity, this globe is to be pitied for having given birth to such creatures, but its value is enhanced by bearing on its surface intelligences capable of rising to the highest contemplations of the spectacle of nature.”

It is with these words that Kant in his *Natural History of the Heavens* terminates the last chapter, which treats of the habitability of the planets. They come from the soul of every thinking man who has preserved the spark of the ideal. The aspect of the starry sky awakens all those who are not among these

vulgar spirits the same sensations which manifest themselves in various ways according to the education of him who experiences them and according to his momentary disposition. The astronomer also is delighted by this spectacle in spite of his regular observation of the sky. The astronomer sees farther and clearer into the celestial spaces than the ordinary man, his knowledge leads him by rapid deductions to vast considerations until he suddenly reaches a point where, for the time being at all events, an insuperable barrier presents itself to the human spirit, and where an imperious halt reminds him of the truth that all science is incomplete. The ordinary man looks upon the sky in quite a different way, and so does a woman. To both, the spectacle of the starry sky offers an enjoyment independent of all research and all preoccupation. A purely æsthetic pleasure.

Are these stars inhabited? Are they inhabited by thinking beings? Do love and hate reign there as they do here? Such are the questions which occur at once. Afterwards, in hours when the need for hope and consolation is felt, the desire arises to contemplate some day with our own eyes the splendours of those other worlds and to be able to soar up to them. The most brilliant stars are the special objects of such wishes, and if they could be fulfilled a population of loving souls would reside on these same stars: Sirius, Vega, Venus, and Jupiter.

The question of the population of the celestial bodies is as ancient as the discovery of the individual existence. It has always occupied thinking humanity, and many have sought to raise the veil which hides the answer. We, also, should like to take a step towards this solution, and it is with that object that we propose to quote here the actual facts bearing upon the conditions of habitability of the celestial bodies and to draw the most probable conclusions. We can combine the astronomical data with the most recent use of physics, and, thanks to the progress recorded by these two sciences in the last ten years, we may be able to draw some new conclusions.

(Here the author passes in review the older writers, Huygens, Kircher, and Fontenelle, which I have summarised in my two books called *The Plurality of Inhabited Worlds and Imaginary Worlds*. He arrives eventually at the nineteenth century and at Gruithuisen, whom my readers know less.)

HISTORICAL

Huygens, the most celebrated mathematician and physicist of his epoch (1629-95), takes it for granted that all the planets are inhabited (he does not speak of the Sun or stars), and that consequently all of them offer the conditions vital and essential to us, that is to say air and water. He easily refutes the objections based upon the distance of the planets from the Sun. He supposes that the water on the other planets will have quite other qualities than ours; on Mercury, for instance, it would only boil at a very high temperature, and on Saturn—at his time the outermost planet known—it would not freeze at the lowest temperature then conceivable. The mass of Jupiter compared with that of our own globe would suggest that its air must be very dense and that its inhabitants could swim in it; but the Jovians would easily accommodate themselves to this state of things. He supposed that the mind of the inhabitants of the other planets was about the same as ours and that their organism was analogous to our own; for what would be the object of the Sun illuminating those other planets if their inhabitants had no eyes? One might indeed think that different species of reasonable beings might exist, but not on the same planet, for they would be in mutual conflict, would struggle for supremacy, and do each other every kind of harm.

He gives a very naïve refutation of an opinion given before him that the height of the inhabitants would have to be inversely as the volume of the planet and consequently the men on Jupiter would not be bigger than the mice on the earth. This, he says, is not possible, because such small beings would not, as astronomers, be capable of using large telescopes.

An important question, according to Huygens, is whether the intelligence of the inhabitants has any relation to their distance from the Sun. He inclines to believe that the inhabitants of Mercury are much more intelligent than we on account of the greater force and vitality of their spirit due to the greater heat of the Sun, but this is not confirmed by what happens on our own globe. The same reason would lead one to believe that the inhabitants of Jupiter are much less intelligent than ourselves, although its four satellites would offer to the mind material for profitable astronomical studies.

What is also evident is the small value of a purely logical and philosophical reasoning based on insufficient premises. We see how a man of sense and judicious spirit can be led to absurd conclusions when he is burdened with preconceived ideas and incomplete knowledge. We shall have several more occasions to bring out this fact, and that is why we have avoided pretending to resolve the question we have put. The deductions which we

formulate to-day may be reversed to-morrow by new discoveries furnished either by experience or by theory.

Apart from his decided opinion concerning the habitability of all the planets, Huygens draws his conclusions logically enough, inasmuch as he bases his affirmations on the knowledge of his time.

An entirely opposite method is used by the Jesuit Father Kircher.

He starts from the point of view that the principal object of nature is man and that all the rest is created for him. The planets are uninhabited because, apart from man, there could be no reasonable beings. But since they have upon man an influence determined by their astrological value, he finds the planets to be such as astrology represents them to be in their action upon us without considering their position with regard to the Sun.

On Mercury, everything takes place gaily and joyously, since all those who are born under its influence are inclined to lightness and mischief. On Venus, everything is even better—or worse: he finds everything gracious and charming; a soft rosy light is spread over the planet, perfumes are wafted about everywhere, zephyrs mingle their murmurings with those of the brooks, and gold and precious stones sparkle everywhere. Jupiter having, like Venus, a beneficent influence upon man, everything there is perfect: the air is pure and wholesome, the waters crystal-clear, and the soil itself as bright as silver. On Mars, on the other hand, everything is of a warlike roughness, forbidding and terrible, rivers of boiling pitch overflow their banks and envelop the country in thick and suffocating smoke. Saturn, as a planet, is particularly accursed; everything looks like a deserted grave. The planets are not inhabited by human beings, but by angels or genii who rule them.

(All these arguments of Kircher are not less infantile than those of Huygens, but they have left distinct traces in astrological literature, and even Victor Hugo reflects them eloquently.)

A contemporary of the two authors we have just mentioned is the Nestor of French writers, Fontenelle, who lived from 1657 to 1757, exactly a century. He described with much detail the inhabitants of the planets, and, like Huygens, he starts from the basis that they are all inhabited, and inhabited by beings formed according to the circumstances. On Mercury, according to him, the heat is so great that the rivers contain fused metals instead of water, particularly gold and silver; the inhabitants of this planet can therefore not imagine that there are worlds like the Earth, where gold and silver are solid

and serve as money. Besides, the inhabitants of Mercury could not support the excessive heat if their planet was not animated by a movement of rotation so rapid that they are only exposed for a short time to the rays of the Sun. They are all inclined to be hot-heads and, like fools and infants, live without reflection and enjoy themselves in anticipation of the coolness of the night. Littrow remarks on this subject that Bode, the translator of Fontenelle and at one time Director of the Berlin Observatory, is seriously astonished by this opinion of Fontenelle and exclaims: "Very strange! for with us in Berlin we find that a great heat makes people lazy and sleepy instead of lively and active."

The inhabitants of Venus only render homage to the goddess of love. They are not interested in philosophy or mathematics, read no books or journals, pass the whole day in their flirtations, and practise in a superior manner the arts which appertain to them, music, poetry, dance, etc., but they are not adept at cookery, for they live almost entirely on air. They are not beautiful, but their amorous character prevents them being influenced by their ugliness; they are Celadons and Sylvanders. Wieland certainly did not know the works of Fontenelle, or he would probably have located one of his romances or his love-stories on Venus as depicted by that author.

Fontenelle's procedure with regard to Mars is rather singular. He declares that that planet does not merit any attention. Our imaginative savant hardly wants to say anything about Jupiter either. He gives a description of the aspect offered by the whole solar system as seen from that planet. He explains how Venus and Mercury are invisible there without the aid of a telescope, and that the Earth only appears as a point. The volume of Jupiter causes him some embarrassment, for whereas the inhabitants of Mercury on account of the small dimensions of that planet nearly all know each other, those of Jupiter cannot possibly do so.

On account of the extreme cold, the life on Saturn is still more disagreeable than that on Jupiter. If the Saturnians were brought to the Earth they would certainly die of heat even in Lapland. If the water on Saturn is of the same nature as ours, it must look like our polished stones, and spirits of wine must resemble diamond. In consequence the inhabitants of Saturn cannot but be slow and phlegmatic; they know no gaiety and remain like oysters in the place they were born in.

Fontenelle continues in this way without attaining any depth, under the impression that our planet is the type of the universe. Let us now pass to the

nineteenth century.

Graithuisen, the Director of the Munich Observatory, published his chief works in the first thirty years of this century. His researches relative to the habitability of the planets were, therefore, made at an epoch when we already possessed important data on their physical constitution, an epoch when, thanks to the work of Bessel, the golden period of astronomy had already begun. One may therefore take it that the work of Graithuisen marks a real progress beyond his predecessors, as indeed he says himself without ceremony. But as a matter of fact he is rather fruitless, as is shown by the strange manner in which he deals with the earth-light on Venus.

He already knew of the phenomenon, which is insufficiently explained even in our day, that during the phases of greatest visibility of Venus the dark side appears to have a faint luminosity. "The simplest explanation to give of this," he says, "is that at the epochs when this faint light on Venus is visible, the inhabitants of the planet organise festivals and general illuminations, which are the easier to arrange on account of the vegetations of Venus being incomparably more luxuriant than even the virgin forests of Brazil. These festivals are probably celebrated on the occasion of political changes or according to religious periods. Now, the principal observations of the ashen light on Venus are those of Mayer in 1759 and of Harding in 1806." Whence he draws the following conclusion: "Between the observation of Mayer and that of Harding 76 years of Venus and 47 Earth years have elapsed. If this period has a religious character, we cannot see a justification for that number of years, but it becomes more comprehensible if we assume that some Alexander or Napoleon then attained universal power. If we assume that the ordinary life of an inhabitant of Venus lasts 130 years of Venus, which amount to 80 terrestrial years, the reign of such an autocrat could easily last 76 years as reckoned on Venus. I have no intention to press this opinion and do not claim its credibility, even should it appeal to the reader's imagination; but if my hypothesis is correct, we at least receive direct testimony to the existence of inhabitants on Venus. Even if the period were shorter, the phenomenon might still be due to some other observance. One could celebrate all the great festivals by similar illuminations which would sometimes follow close upon one another. The fires would serve another purpose, inasmuch as they would thin out the forests and provide fresh arable ground for an increasing population. Large migrations of people would be prevented, and the consequent wars would be avoided and the race would remain united."

One must acknowledge that these ideas of Gruithuisen are most fantastic. I have frequently observed this unilluminated hemisphere of Venus, notably in September 1895 and in April 1897: it appeared to me of a violet colour, and the idea of illuminations by the inhabitants is pure romance.

He also then passes in review the ideas of Kant which one finds expounded in the two works cited above and which therefore I need not quote. We know that for the Philosopher of Königsberg the intelligence and the degree of perfection of the inhabitants of the planets is in proportion to their distance from the sun.

We may add to the remarks of M. Scheiner that the writers who have dealt with the question of the plurality of worlds have nearly all judged the planets from the appearances they present to us from our point of observation, and have assumed the harmony of nature according to the manner of Bernardin de St. Pierre. For this occasion I have re-read his book with some interest. To this very simple member of the Institut, Venus is a bright world peopled by amorous natives “who give themselves up to dance, festivals, and songs, or compete for swimming prizes, like the happy islands of Tahiti”; the inhabitants of Mars are warlike, “resembling the northern Germans, their forests and hills, their atmosphere resounding to the warlike sound of their horns and that of drums and trumpets which announce the spilling of blood”; on Jupiter “they resemble the Dutch, being industrious, patient, wise, reflective, and tending their numerous herds in their vast fields,” etc., etc.

The author of *Paul et Virginie* remains purely terrestrial in these descriptions, which he believes to be astronomical.

But let us return to the dissertation of M. Scheiner. He passes on to the purely scientific aspect of the question, which is the only one which interests us here.

THE ORIGIN OF LIFE

The problem of knowing if worlds other than the Earth, being habitable, are really inhabited, depends really upon this question: How did life appear on this Earth?

It is irrefutable that there was a time when the Earth was not habitable in the usual sense. Therefore life necessarily had a commencement. “This can have taken place in three different ways, either by a special act of creation, in

which case it is of little importance to our problem to know whether this act was accomplished in a complete manner as taught by the Bible or whether it was limited to the creation of the inferior forms of life; or by spontaneous generation; or, lastly, by the importation from space, in which we can just as well imagine germs of life as gases and inorganic substances.

“From the philosophical point of view, these three hypotheses are equally well-founded, for none is more easily conceived than the other and observation has not yet confirmed any of them.

“If we admit the first hypothesis, a creative act, such a manifestation of the will of a Supreme and Impenetrable Being is beyond the laws of nature. In this case we have no means of reasoning on the purpose of the Divine Will. We do not know if the creative act has been exercised duly once in favour of the Earth, or whether it is renewed on various occasions, or whether it takes place in the same manner every time a heavenly body becomes fit to act as an abode of living beings. We cannot express any opinion on that and therefore our initial question remains open. In the narrow Biblical sense the Earth and mankind must be considered the last word in creation and the idea that other reasonable beings might exist is necessarily eliminated.” It seems to me that in our times none of the readers of this book need consider the first hypothesis as admissible. Everything shows us that living beings have not been directly created by a Supernatural Will, but that they have slowly and gradually evolved during the geological periods which are known to us. Scheiner then passes on to the second hypothesis. “By spontaneous generation,” he says, “is meant the formation by material molecules of an organism of the most rudimentary species, which involves our attributing to an inorganic substance properties which end in the production of life.

“The adoption of these properties is possible in two ways: by a sudden coincidence of favourable circumstances, or by a continuous process which, thanks to a gradual development, fills up the gap of continuity which at present appears to exist between inert matter and living matter.

“If circumstances on our earth have favoured spontaneous generation, there is no reason why they should not have done the same on other celestial bodies having a similar constitution, and one may deduce with certainty that all the heavenly bodies which are in this condition are provided with similar organisms.” The author then examines the third hypothesis.

Space can be filled with organised matter or matter capable of life distributed almost uniformly, without being specially destined for any particular body, since we must assume a commencement for each. The surface of the heavenly bodies receives the organisable matter which, when it finds the necessary conditions, develops and forms living beings. It is clear that in this case the presence of organised beings on all the heavenly bodies capable of entertaining life is not merely a probability but a certainty.

Scheiner considers the three hypotheses as equally acceptable. The first, he says, is a matter of sentiment. It does not solve the question, since it would involve a Divine Will. Out of the three hypotheses the last two would solve our problem in an absolutely affirmative manner, whereas the first leaves it undecided.

“We only wish to prove one thing by our argument, and that is that the opinion that habitable heavenly bodies are really inhabited is much more probable than the contrary opinion, and this authorises us to continue to develop our thesis.”

THE CONDITIONS OF LIFE

The author here puts a fundamental question. What is life and what is living matter! Libraries have been written on this subject but all to no purpose, for we are as unable to seize the essence of life and of living matter as that of gravitation, for instance, though the latter appears to us infinitely simpler in its manifestations than what has been called vital force.

We only know that on our Earth vital force is united to a special form of matter called organised matter, and that when this organised matter disappears it ceases to exist as vital force and transforms itself—since energy is indestructible—into other forms of energy. It is not of material importance whether vital force is regarded as a special force or a special aspect of a known force, *e.g.* electricity. It follows from this intimate association of vital force with living matter that the vital manifestations (nutrition, growth, reproduction, etc.) can only take place under conditions in which living matter can exist; in all other cases the manifestations of life cease and death ensues, or the vital force becomes latent until favourable conditions return.

Our problem of the habitability of heavenly bodies is therefore limited to the question of celestial bodies on which conditions are such that living matter can exist in a permanent form. We must therefore enquire first of all what

those conditions are, after which we shall be able to use our astronomical resources to find whether these conditions are represented on the other centres of condensation of matter in the universe.

The conditions necessary to life are the more numerous the more complicated the structure of organic matter. The maximum of requirements is therefore attained in the higher animals and in man.

The simpler the organism, the simpler are its conditions of existence and the greater are in general the possibilities of supporting unfavourable conditions.

Animals and plants living in caves or at great depths under water are deprived of light; they have accommodated themselves to that privation and do not suffer by it. Animals require oxygen in air and in water; plants also require a small quantity of carbonic acid for building up their tissues. There are even animalcule in existence for whom oxygen is a poison. As a general rule, temperatures above 50 degrees centigrade are insupportable.

This is due to the fact that at that temperature albumen, one of the most important substances in the animal organism, coagulates. Inferior beings can resist higher temperatures and even for a short time 100 degrees centigrade, which is the boiling-point of water, but they could not live long. In its liquid form water is indispensable to organic life; life of any duration below zero is impossible because the water contained in every organism would solidify and the parts composing the organism would lose their mobility. Lack of water does, however, not inevitably imply death, and plants particularly can preserve for a long time a latent vitality while deprived of water. Cereals furnish a striking example, for when dried they can preserve their germinating power for years. Although living matter can preserve its vitality so long, it is none the less true that during that period all manifestations of life cease. If therefore the lack of water is perpetual, life must be considered as really suppressed.

Even for the lowest forms of life three conditions must be regarded as essential: water, an atmosphere containing oxygen and carbonic acid, and a temperature between the limits indicated above.

It is therefore really from these three points of view that we must study the heavenly bodies in order to be in a position to judge whether organic life as we know it is possible on them or not. As regards knowing whether that life presents itself under forms comparable to those which we see here, whether

there are, for instance, beings analogous to humanity, that is quite another question.

The means at the disposal of astronomy for determining the constitution of the heavenly bodies are of various kinds. We can take into account phenomena which at first sight do not seem adapted to that end.

Direct observation with the help of the telescope enables us to discover the surface details of the planets and any changes which take place in them. Such changes in most cases imply the existence of an atmosphere. Observations of occultations of stars by the Moon or the planets lead to the same result. Theoretical astronomy tells us the distance between the planets and the Sun; physics tell us the quantity of light received by each planet from the Sun; and the period of revolution and the inclination of the planet's axis tell us about the course of its seasons. Photometry gives us the amount of sunlight reflected by the surface of the planet, and thus furnishes indications concerning certain properties of the planetary surface, properties which permit us to decide with certainty, for instance, whether light is reflected by a solid surface such as the ground or whether the rays do not penetrate so far and are sent back in the upper parts of the atmosphere by layers of clouds.

It is spectrum analysis which furnishes, as we know, the most important auxiliary information; it presents the heavenly bodies to the eye of the mind as the microscope unveils to the eye of the body the marvels of the infinitely small. The rays of light are messengers who, having passed through the spectroscope, bring to us news of the most distant worlds and tell us of the temperature of the fixed stars, of the metals volatilised in their atmospheres, of the incredibly low temperature of the nebulae and of the gases which envelop the planets.

We do not here wish to intone a hymn to spectrum analysis; we only wish to report briefly and simply what we know of the physical nature of the celestial bodies. But we must admit that the greatest amount of that knowledge is due to the spectroscope.

THE PLANETS OF OUR SYSTEM

(We continue the translation of the essay of the Potsdam astronomer.)

In the light of contemporary astronomical knowledge let us make a rapid survey of the other worlds.

The Moon

Our lady readers will not raise any objection if we occupy ourselves at first with the Moon, this confidante of every heart, either happy or unhappy. The most important thing about her for the moment can be put into one sentence: she has neither air nor water, and her temperature oscillates between extremes separated by more than 200 degrees centigrade. None of the conditions stated above is therefore fulfilled, and accordingly no organic life could exist on her. It is also interesting to see how the cessation on the Moon of one of these vital conditions has entailed the cessation of the two others. There is no occasion to doubt that the Moon formerly possessed an atmosphere; by analogy with the planets this must indeed be assumed as altogether certain. The feeble mass of our satellite which, on the one hand, has been the cause of its rapid cooling, has, on the other hand, brought about the dissipation into space of its atmosphere, which was probably always of small density.

But the smaller the pressure of the air the more rapid is the evaporation of water, and this is why its disappearance coincides with that of the atmosphere. Besides, the complete absence of air allows the rays of the Sun to penetrate without hindrance to the ground and to heat it to a high temperature during the 14 times 24 hours of duration of the lunar day. During the night, which is of equal duration, there is radiation of heat into celestial space, and the soil cools down to a temperature which cannot be very different from the absolute zero of the temperature of empty space.

(I cannot entirely accept these allegations with regard to the habitability of our satellite. The absence of air and even of water is not proved. The variations actually observed even prove that the moon is not altogether a dead world. But let the author continue.)

Such a fate is also in store for the Earth, and nothing can save it. Our Earth will also one day become a barren body, incapable of supporting organic life, a deserted grave of the civilisation created by the human spirit. Just as the isolated individual disappears, so will humanity one day disappear entirely. In a limited domain death is always finally victorious; but, on the other hand, a new life flourishes elsewhere, and when that new life is developed, some day perhaps on another planet of our system a scientific article will be written on the question whether the Earth is still habitable.

Mercury

Our knowledge of the physical constitution of Mercury is very slight. It seems to be surrounded by a light atmosphere which contains water-vapour. Since the solar heat on Mercury is about seven times stronger than on us, the extreme limits of temperature above indicated must here be very considerably surpassed, and water can hardly exist except in the form of vapour. But the argument is entirely changed if we accept the recent discovery of Schiaparelli, according to which the duration of rotation of Mercury would be identical with its period of revolution, so that, like the Moon to the Earth, Mercury would always turn the same face to the Sun. On that side the temperature would naturally be very high, while the most intense frost would reign on the other side. But between these two extremes there ought to be a mixed zone in which our three conditions might possibly be realised, so that we may assume for Mercury a limited habitability.

Venus

On Venus the Sun's heat is still very considerable, and in the torrid zone it would be insupportable to us. But, on the other hand, this planet is surrounded by a dense atmosphere, which on account of the presence of water-vapour proves the existence of water on the planet. The higher regions of the atmosphere are occupied by a thick layer of clouds which hardly ever allows our gaze to reach the ground, but which is equally opposed to the passage of the Sun's rays. Much more than half the solar radiation is reflected by that layer of cloud, and we can suppose, on the whole, that the upper limit of a tolerable temperature is not passed at the surface of this planet. And as we have already said, since there is water and an atmosphere, we have no reason to doubt the habitability of Venus.

Mars

As regards Mars, which commences the outer series of planets, we obtain a still more satisfactory result. We can clearly recognise the subdivision of the surface into water and dry land. Its atmosphere has properties which agree with those of our atmosphere. Not only is there certain evidence of the existence of water-vapour, but spectroscopic studies have proved also that the principal components of the atmosphere are the same as those of the Earth's atmosphere—that is to say that there is oxygen and nitrogen. Sometimes the soil is hidden by groups of clouds, sometimes they disappear to appear again in

other places. Its poles are encased in snow and ice, the white area of which varies in extent according to the seasons. Besides a number of enigmatical facts—we need only refer to the canals and their doubling—there are numerous meteorological phenomena on Mars which frequently occur in our own atmosphere. Although the temperature of Mars is, on account of its farther distance from the Sun, sensibly below that of the Earth, the difference is not sufficiently considerable to oppose a serious obstacle to the stable existence of organised matter in the torrid or temperate zone. The torrid zone of Mars must correspond approximately in climate to our temperate zone. We can therefore finally declare with entire conviction that the conditions offered by Mars are suitable for life as we know it on Earth.

Jupiter, Saturn, Uranus, and Neptune

With the planets Jupiter, Saturn, Uranus, and Neptune we enter a region quite different from the preceding one. These bodies all possess atmospheres of great density in which, on Jupiter for instance, we can observe immense revolutions and violent cataclysms. They also contain water-vapour, but besides that they contain a substance characterised by a strong absorption of certain red rays. This gas is found in a small quantity on Jupiter, but in much greater quantities on Saturn and Uranus. On the latter planet the atmosphere, apart from the water-vapour contained in it, does not seem to have any analogy with ours; strong absorption-bands show in the less refrangible portions of the spectrum. As we have seen that oxygen is not absolutely necessary to organic life, the single fact that the outer planets have an atmosphere may suffice to show that our first condition is fulfilled. The presence of water-vapour in their atmosphere proves the existence of water and fulfils the second condition. As regards the third condition, Jupiter also seems to satisfy that, at least in equatorial regions, especially if we take into account that radiation of heat into space is much limited by the thick atmosphere filled with clouds. The farther we go away from the Sun the more does the third condition become precarious, and while we may have some doubt concerning Saturn, it cannot be denied that on Uranus and Neptune the solar heat is insufficient to support organised life in a durable way.

* * * * *

(The atmospheres might be formed of gases which would make the radiation almost zero and would produce a relatively high temperature. This

certainly happens in the case of Mars, whose temperature is not lower than that of our globe, where the polar snows are less dense than ours and melt more completely in the course of the summer.)

* * * * *

But another peculiarity presents itself which may reverse all our ideas relative to the planets from Jupiter outwards. Certain observations tend to show that Jupiter is not yet cooled down, that its real nucleus is still fused or perhaps even in the gaseous state, and that it has not yet formed a solid crust upon which life might develop. Besides the phenomena directly observed or revealed in the spectroscope, the very small specific gravity of these planets also supports that hypothesis—the density of Saturn is about the same as that of cork. It is very difficult to form an exact idea of the constitution of these planets. On the other hand, it is possible to believe that one of the extreme planets, Uranus for instance, is sufficiently cooled to possess at least a liquid surface which, on account of the internal heat of the planet, may have preserved for a certain period a temperature sufficient to entertain life even after the solar heat is no longer sufficiently powerful. But these are only hypotheses.

* * * * *

Let us summarise in a few words the results of this chapter.

On the Moon no organic matter can exist; we may suppose living beings to exist in a small zone of Mercury; the surface of Venus is very probably habitable in most of its regions; Mars is certainly habitable and probably under such conditions that certain species of our plants and animals could, if transported to that planet, continue to live there.

For the other planets the possibility of habitation cannot be entirely denied, but the existence of living beings on their soil is not probable. Let us add, for the sake of completeness, that on the Sun, which radiates the most intense heat, and on those innumerable other suns which we call stars and which the telescope reveals to us, living matter certainly does not exist.

It follows that of the millions of stars visible to us in the universe there are only two or three which we could consider with any certainty capable of being inhabited as we conceive it. That seems a rather unsatisfying result, which

allows an icy sentiment of loneliness in infinite space to take possession of our souls.

We have seen in the first part of our work how the speculations of those who formerly discussed the habitability of the planets not only raised the question whether such and such a planet was inhabited, but also tackled the much vaster problem of the particular characters of the beings which lived there. After attentively following the discussion, nobody should ask to penetrate further into this question. It has indeed been a great effort to obtain a positive result at all, even while restricting the population of the planets to the simplest forms of organised matter. The number of forms under which it shows itself on the Earth is so considerable that we can only be dazed by its abundance, and this impression must be even increased if we remember that the possible forms are far from being exhausted and that we only know those which, satisfying terrestrial conditions, have been able to survive by adapting themselves to their surroundings. We do not exaggerate in declaring that Nature recognises no limit to the number of forms which can harbour life, and this observation is an argument in favour of those who desire to find reasonable or superior beings in distant worlds. It indicates that as soon as the first germs of life exist, the possibility of a complete development is there, and that however different the external conditions may be, Nature is not at a loss for varieties of forms of life. That is why we have the right to hope that on Mars, for instance, there are beings who not only show manifestations of animal life, but who are endowed with intellectual faculties. That, however, is all we can acknowledge: the right to hope and not to a certainty.

Yet at the risk of exposing ourselves to a criticism similar to that which we have in the first part directed against other authors, we would now invite the reader to follow us for a moment into the domain of speculation and hypothesis.

THE POSSIBILITY OF BEINGS OF DIFFERENT CHEMICAL CONSTITUTION

In this chapter Scheiner develops the idea indicated above (page 127) on the possibility of the existence in other worlds of living beings entirely different from ourselves.

We have up to now (he writes) understood by organised matter something of which *carbon*, combined with hydrogen, nitrogen, and other elements, is the

principal chemical component. Carbon is the essential constituent; organic chemistry is the chemistry of carbon. We do not know any other substance which would allow of such an array of combinations, but the possibility of such a body cannot be denied. While on Earth all life is bound up with carbon compounds, one may suppose that in entirely different circumstances another element might show itself capable of supporting the conditions of life in combinations which might resist greater heat without decomposition or greater cold without becoming torpid. A few years ago we seemed to be on the track of something like that. *Silicon* is the element which has the greatest chemical analogy to carbon, and in combination with oxygen it is found in enormous quantities in the form of silica, and all its combinations have very characteristic properties. Just as in organic matter every being forms itself into cells by fission or conjugation, so also can we produce from combinations of silicic acid a cell from which under our eyes an object develops which has a vegetable appearance. This experiment, easy to make, is perhaps unknown to many of our readers, and therefore we must say a little more about it.

Most combinations of silicic acid with metals are insoluble in water. Some, however, and especially a combination with potassium known by the name of water-glass, occurs in a soluble form. If into such a solution we introduce a small quantity of any soluble metallic salt such as chloride of copper or silver nitrate, decomposition takes place and the chloride of copper is converted into silicate of copper and chloride of potash. A very curious phenomenon then takes place. At the moment when the first traces of the metallic salt are dissolved, the decomposition mentioned above takes place and the molecule of salt is covered with an extremely fine skin of insoluble silicate. It is across this skin that the liquids are then exchanged by a process of osmosis which, as we know, plays a great part in the life of plants and animals and which here takes place as follows:

More liquid enters the membrane than issues from it; consequently the internal pressure increases until it is great enough to break through the membrane, somewhere allowing a drop of the liquid which is a solution of metallic salt to escape. On account of the chemical decomposition, this drop immediately surrounds itself with a new skin, and the process of rupture and reformation is repeated until the metallic salt inside the membrane is completely decomposed. We thus see arising under our eyes in a few minutes a marvellous arborescent structure which could at first sight be taken for some plant of inferior order. The rupture of the membrane always takes place in the most recently formed portion, because there the membrane, which thickens

gradually, has as yet the least thickness; that is generally the upper portion, so that this offers another analogy to plant life. The coloration of the cells differs with the nature of the metallic salt, but we cannot trace any influence upon the forms obtained. By throwing into the solution several different metallic salts, one can form a many-coloured garden in a small bottle.

We are far from seeing in this experience anything but a quite external resemblance to the phenomena of living matter. Yet it leads us to other reflections, especially as we do not yet know anything concerning the real nature of living matter.

Among plants and inferior species of the animal world we cannot imagine a life conscious of itself. But in common with our silica creature they show growth and the consumption of chemical substances. In both cases that growth is arrested wherever and whenever the food gives out. This happens in the silica solution when the provision of metallic salt is exhausted; and these are not the only analogies which one can find.

But it will be said that there is in reality a capital difference: in one case we have to do with real life for which we have no explanation; in the other we have to do with a simple chemico-physical phenomenon.

We reply that a few centuries ago this chemical vegetation would have been an enigma as life itself is still to us, and nobody would then have doubted that he was observing the development of some strange plant. Would they not then have taken the silicon cell for an organic cell? What follows? That the idea of living nature is quite relative, that it changes with our knowledge, and that an imaginative spirit is quite at liberty to endow the stare which we just excluded from life with a life quite different from ours.

Apart from these possibilities, the somewhat discouraging result of our previous study was that of all celestial bodies visible to us only two or three could be described with any probability as fit to support something resembling our terrestrial organic life. We cannot object to this conclusion so long as we take it literally, but we wish to point out that all depends upon the little restriction contained in the words "visible to us."

We must therefore include the heavenly bodies which we cannot see and concerning which we know very little. But here we encounter a very peculiar paradox. We know nothing about these invisible stars, and yet as far as our interests are concerned we know more about them than the others. For

mathematics comes to our aid, and if we rely on the calculus of probabilities we arrive, as we shall see, at very clear results.

Our Sun has created for itself, without counting the asteroids, a retinue of eight planets, which on account of their respective distances from the central body are placed in the most diverse conditions as regards temperature. Out of these eight planets one, the Earth, is undoubtedly inhabited, and two others, Mars and Venus, very probably. From the fact that the Sun has produced not one planet but eight, we can conclude that very probably the other suns or fixed stars have also produced one or several planets, and that those which escape from that law are the exception. We must also admit that among these stellar planets there may be some in such a condition and at such a distance from their central sun that organic life is possible on their surfaces.

We shall make this calculation with figures so modest that we shall obtain results obviously below the truth.

The number of stars revealed by a telescope of moderate power amounts to 10 millions. If we suppose that every star only has on an average one planet, we obtain already the considerable number of 10 million planets. It is true that with us three planets out of eight may be considered habitable, but let us suppose the proportion in the universe is only one in a hundred. We shall still have no less than a hundred thousand habitable planets.

This number, evidently falling short of reality, makes already quite another figure compared with the three habitable worlds we had before.

After that the universe no longer appears such a desert. There is nothing to hinder us from giving rein to our fancy and imagining on the one hand the strangest forms of life among the innumerable planets gravitating round the stars, and on the other hand thinking beings which surpass us considerably in intelligence and to whom our most difficult problems are as transparent as self-evident truths. The conclusion is that in all sorts of degrees we must see Life radiating into space and lighting up Infinity.

REMARKS

To this interesting study of Monsieur Scheiner we may add in the first instance that the question is still more complex, that we can consider the resources of Nature as infinite, and that “positive” science founded only upon

our senses is quite insufficient, though it may be the only basis of our reasoning. It is through the eyes of the spirit that we must survey the universe.

As we have seen, this new examination of the question of the habitability of other worlds by thinking beings presents a new interest, and the author has been able to escape the usual error of scientific writers, which consists in supposing that the first condition of habitability of another world is that it resembles the Earth. That is always the reasoning of the fish, which would affirm without an absolutely logical conviction irrefutable to himself that it is impossible to live outside water. But it seems to us that our conception of the universe can be even more vast and elevated than that of the learned German astronomer.

As regards the planetary systems differing from ours, we are no longer obliged to fall back upon suppositions. We know already with certainty that our Sun is no exception, as some theoreticians maintained even quite recently. The discovery is of considerable interest.

It is surely a rather exceptional situation that a sidereal system consisting of a central sun and one or more bodies gravitating round it should have its system just in our line of vision so that the revolution of the bodies which compose it should bring dark bodies between us and the star and produce a more or less complete eclipse. Since, on the other hand, such eclipses would be our only means of proving the existence of these unknown planets (except perturbations as in the case of Sirius and Procyon), it seems that it would have been absolutely daring to hope for such a circumstance to discover solar systems different from ours. Yet this exceptional case occurs in various parts of the sky. Thus, for example, the variable star Algol owes its variation of brightness, which reduces it from the second to the fourth magnitude at intervals of 69 hours, to the interposition of a body between it and the Earth, and celestial mechanics has already been able to determine with precision the orbit of this body, its dimensions and its mass, and even the ellipticity of the Algol sun. Thus we have here a system of which we know the sun and one enormous planet whose revolution takes place in 69 hours with a very high velocity as measured in the spectroscope, a planet still self-luminous, although less luminous than its sun, as the Earth was long ago, but a planet in the course of cooling and approximating to the state of Jupiter.

The star Delta Cephei is in the same case: it is an eclipsing variable with a period of 129 hours, and its eclipsing planet also revolves in the plane

containing our line of vision. The star U Ophinchii shows a similar system, and observation has revealed several others.

If therefore chance has brought it about that a certain number of different solar systems should have been revealed to terrestrial observation by presenting side views, that is evidence of the existence of innumerable solar systems disseminated through the depths of space, and we are no longer reduced to mere conjectures.

On the other hand, the analysis of the movements of several stars, such as Sirius, Procyon, Altair, and many others, proves that these far-distant suns have companion planets as yet unrevealed by the telescope, and which possibly may never be discovered because they are dark and lost in the radiation of the star. The companion discovered in the neighbourhood of Sirius is not the only celestial body of that system. Scheiner speaks of about 10 million stars as constituting the sidereal universe. But the photographic chart of the heavens which comprises stars down to the thirteenth magnitude is already expected to contain 30 millions. If we go down to the lowest magnitudes we reach the figure of 100 millions. It is therefore not an army of 100,000 worlds which appears before us, but rather of several millions.

Now, this is a point of the greatest importance for the exact appreciation of the problem.

The terrestrial organisms from the lowest up to man are the result of forces in action on the surface of our planet. The first organisms seem to have been produced by combinations of carbon with hydrogen and oxygen, and their life consisted, so to speak, only in a few rudimentary sensibilities. Sponges, corals, polypi, medusæ, give us an idea of these primitive beings. They were formed in the warm waters of the primary ages. While there were yet no continents nor islands emerging from the universal ocean, there were no air-breathing organisms. The first aquatic beings were succeeded by amphibia and reptiles. Afterwards came the mammalia and the birds. The constitution of beings stands in close relation with the substances of which they are composed, the medium in which they live, the temperature, the light, the density, gravitation, length of day and night, seasons, etc.—in a word, all the cosmographic elements of the planet.

If, for example, we compare two such worlds as the Earth and Neptune, which differ very much as regards distance from the Sun, we cannot imagine for a single instant that the organic forms should have had the same

development. The average temperature must be much lower on Neptune than on the Earth, and so must the intensity of light. The years and the seasons are 165 times longer than with us; the density of materials is three times less, and gravitation, on the other hand, is a little stronger. Under conditions so different from ours, the activities of Nature can only have been shown in other forms. The elements also are not found present in the same proportions, and spectrum analysis has even shown us that substances which prevail in the atmospheres of Jupiter, Saturn, and Uranus, as well as Neptune, are different from those which constitute our organisms. Lungs functioning in another atmosphere would have to be different from ours. The same applies to the stomach and the digestive organs. Chemical constitution is not even the same. Instead of carbon as a fundamental element associated with hydrogen, oxygen, and nitrogen, we can imagine with Scheiner silicon and other bodies. We must conclude that the organs and the senses cannot be the same as they are here. The optic nerve, for instance, which has been formed and developed here from the rudimentary organ of the trilobite to the marvels of the human eye, must on Neptune be incomparably more sensitive than it is in our blinding sunlight and must perceive radiations which we do not perceive here. It may even be replaced by another organ. The bodily forms, animal and human, can resemble nothing on earth.

Certain *savants* object that, if the conditions are too different from terrestrial conditions, life cannot exist at all. But we have no right to limit the powers of Nature by the narrowness of our sphere of observation, or to pretend that our planet and our human race are the model for all planets. That is an hypothesis as infantile as it is ridiculous.

Others go still farther, and imagine that life only appears on Earth, and that we have no sufficient reason to suppose that on other globes it has been the result of inorganic evolution. This, as we have often repeated, would be a strange interpretation of the language of Nature, considering that our small planet is too small a cup to contain the whole of life, that that life abounds everywhere, fills the waters, swarms in the air, covers the entire surface of the globe, and that the fertility of Nature is such that she multiplies parasitic life at the expense of life itself, rather than get tired of producing. And the spectacle is the same for all the immense duration of the geological eras. Quite lately, noticing a heap of fossils of the secondary era by the roadside in the country, I took a stone to put it into a collection, for it was entirely made up of shells petrified and cemented in a block. In taking it up I exposed a swarming mass of living beings, small snails, wood-lice, beetles; and I caught two lizards;

while butterflies laid their eggs in the plants around. Life of former days, life of to-day, life everywhere. Life always!

Certain minds are capable of supposing that for the whole duration of its existence, for millions of years, a world could come to nothing but the state of dead and barren rock and that the life which swarms on the surface of our planet is only a freak due to the fortuitous combination of elements of fruitfulness, a parasitism of more or less large fleas which might never have been produced at all. But that is a hypothesis contrary to the observation of Nature, and difficult to maintain seriously except as a pure play of the imagination, which does not satisfy the most elementary logic. And though our logic may not be that of Nature, yet we must not stray too far away from it if we want to reason.

But as we have already repeated so often in this book, and what we must thoroughly absorb, is the importance of time as well as of space. Just as our world is nothing but a small island, a point in the universe, so also our era is nothing but a moment in eternity. The present moment has no more importance than the moments which have preceded or those which are to come. There is no reason to believe that such and such other worlds are inhabited at present simply because we live at present and can observe them. One world has been inhabited in the past, another will be inhabited in the future. Let us not be personal, like infants or the aged, who see only their own room. Let us know how to live in the infinite and in the eternal.

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THE END

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